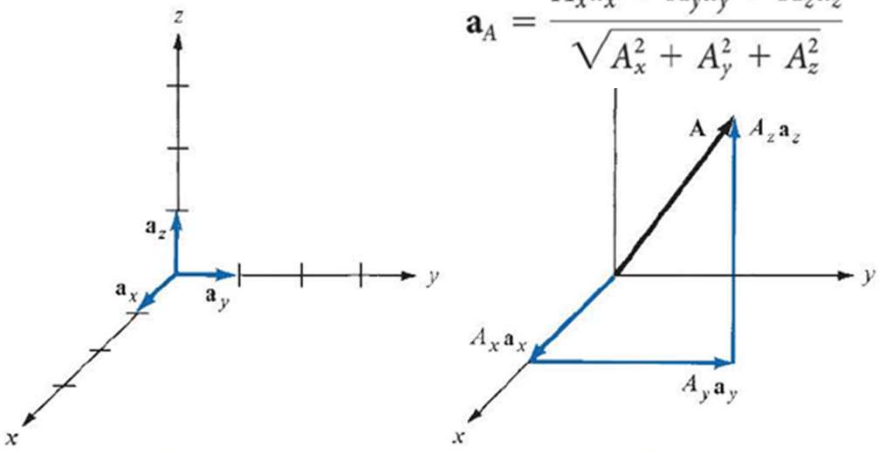
Scalar and Vectors

- A **scalar** is a quantity that has only magnitude.
- Quantities such as time, mass, distance, temperature, entropy, electric potential, and population are scalars.

- A **vector** is a quantity that has both magnitude and direction.
- Vector quantities include velocity, force, displacement, and electric field intensity
- EM theory is essentially a study of some particular fields.
- A field is a function that specifies a particular quantity everywhere in a region.
- Scalar Field(Potential) and Vector Field(Electrical field)

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

$$\mathbf{a}_A = \frac{A_x\mathbf{a}_x + A_y\mathbf{a}_y + A_z\mathbf{a}_z}{\sqrt{A_x^2 + A_y^2 + A_z^2}}$$



Vector and unit vector

- A vector **A** in Cartesian (or rectangular) coordinates may be represented as

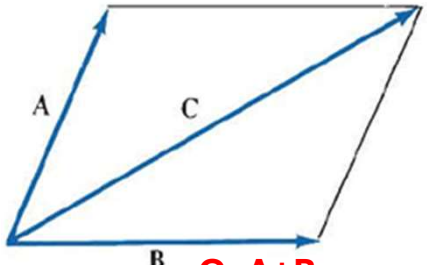
$$(A_x, A_y, A_z) \quad \text{or} \quad A_x\mathbf{a}_x + A_y\mathbf{a}_y + A_z\mathbf{a}_z$$

- A unit vector $\mathbf{a}_A$ along **A** is defined as a vector whose magnitude is unity (i.e., 1) and its direction is along **A**

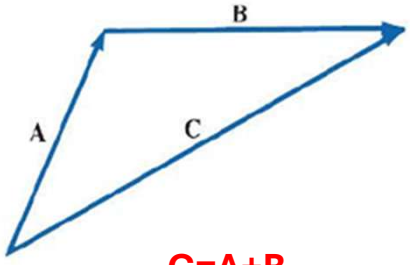
$$\mathbf{a}_A = \frac{\mathbf{A}}{|\mathbf{A}|} = \frac{\mathbf{A}}{A}$$

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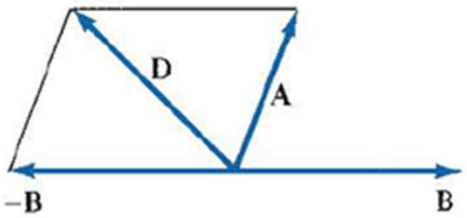
Vector Addition, Subtraction, Position Vector



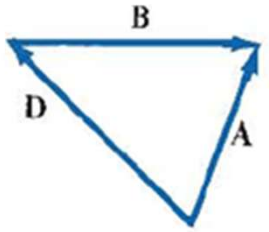
C=A+B



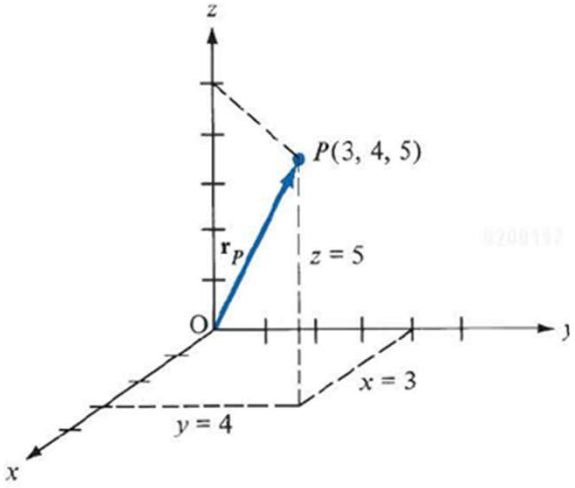
C=A+B

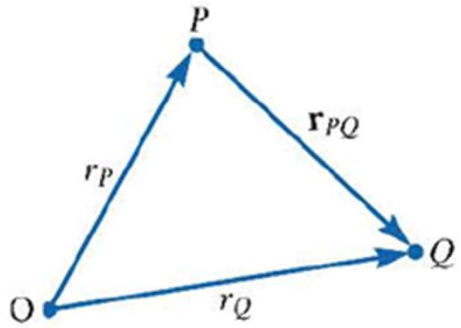


D=A-B



D=A-B





$r_{PQ}=r_Q-r_P$

Points P and Q are located at (0, 2, 4) and (-3, 1, 5). Calculate

(a) The position of vector r_p

(b) The distance vector from P to Q

(c) The distance between P and Q

(d) A vector parallel to PQ with magnitude of 10

Position Vector (contd.)

Given points P(1, -3, 5), Q(2, 4, 6), and R(0, 3, 8), find
(a) the position vectors of P and R,
(b) the distance vector r_{QR},
(c) the distance between Q and R.

Answer: (a) $\mathbf{a}_x - 3\mathbf{a}_y + 5\mathbf{a}_z$, $3\mathbf{a}_x + 8\mathbf{a}_z$, (b) $-2\mathbf{a}_x - \mathbf{a}_y + 2\mathbf{a}_z$, (c) 3.

$3\mathbf{a}_y + 8\mathbf{a}_z$

If $\mathbf{A} = 10\mathbf{a}_x - 4\mathbf{a}_y + 6\mathbf{a}_z$ and $\mathbf{B} = 2\mathbf{a}_x + \mathbf{a}_y$, find (a) the component of $\mathbf{A}$ along $\mathbf{a}_y$, (b) the magnitude of $3\mathbf{A} - \mathbf{B}$, (c) a unit vector along $\mathbf{A} + 2\mathbf{B}$.



Solution:

(a) The component of $\mathbf{A}$ along $\mathbf{a}_y$ is $A_y = -4$.

(b) $3\mathbf{A} - \mathbf{B} = 3(10, -4, 6) - (2, 1, 0)$
$= (30, -12, 18) - (2, 1, 0)$
$= (28, -13, 18)$

Hence,

$$|3\mathbf{A} - \mathbf{B}| = \sqrt{28^2 + (-13)^2 + (18)^2} = \sqrt{1277}$$
$$= 35.74$$

(c) Let $\mathbf{C} = \mathbf{A} + 2\mathbf{B} = (10, -4, 6) + (4, 2, 0) = (14, -2, 6)$.

A unit vector along $\mathbf{C}$ is

$$\mathbf{a}_c = \frac{\mathbf{C}}{|\mathbf{C}|} = \frac{(14, -2, 6)}{\sqrt{14^2 + (-2)^2 + 6^2}}$$

or

$$\mathbf{a}_c = 0.9113\mathbf{a}_x - 0.1302\mathbf{a}_y + 0.3906\mathbf{a}_z$$

Note that $|\mathbf{a}_c| = 1$ as expected.

Position Vector (Problem)

Points P and Q are located at (0, 2, 4) and (-3, 1, 5). Calculate

(a) The position of vector r_p

(b) The distance vector from P to Q

(c) The distance between P and Q

(d) A vector parallel to PQ with magnitude of 10

Solution:

(a) $\mathbf{r}_P = 0\mathbf{a}_x + 2\mathbf{a}_y + 4\mathbf{a}_z = 2\mathbf{a}_y + 4\mathbf{a}_z$

(b) $\mathbf{r}_{PQ} = \mathbf{r}_Q - \mathbf{r}_P = (-3, 1, 5) - (0, 2, 4) = (-3, -1, 1)$
or $\mathbf{r}_{PQ} = -3\mathbf{a}_x - \mathbf{a}_y + \mathbf{a}_z$

(c) Since $\mathbf{r}_{PQ}$ is the distance vector from P to Q, the distance between P and Q is the magnitude of this vector; that is,

$$d = |\mathbf{r}_{PQ}| = \sqrt{9 + 1 + 1} = 3.317$$

Alternatively:

$$d = \sqrt{(x_Q - x_P)^2 + (y_Q - y_P)^2 + (z_Q - z_P)^2}$$
$$= \sqrt{9 + 1 + 1} = 3.317$$

(d) Let the required vector be $\mathbf{A}$, then

$$\mathbf{A} = A\mathbf{a}_A$$

where $A = 10$ is the magnitude of $\mathbf{A}$. Since $\mathbf{A}$ is parallel to PQ, it must have the same unit vector as $\mathbf{r}_{PQ}$ or $\mathbf{r}_{QP}$. Hence,

$$\mathbf{a}_A = \pm\frac{\mathbf{r}_{PQ}}{|\mathbf{r}_{PQ}|} = \pm\frac{(-3, -1, 1)}{3.317}$$

and

$$\mathbf{A} = \pm\frac{10(-3, -1, 1)}{3.317} = \pm(-9.045\mathbf{a}_x - 3.015\mathbf{a}_y + 3.015\mathbf{a}_z)$$

Vector Multiplication

1. Scalar (or dot) product: $\mathbf{A} \cdot \mathbf{B}$
2. Vector (or cross) product: $\mathbf{A} \times \mathbf{B}$
3. Scalar triple product: $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$
4. Vector triple product: $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$

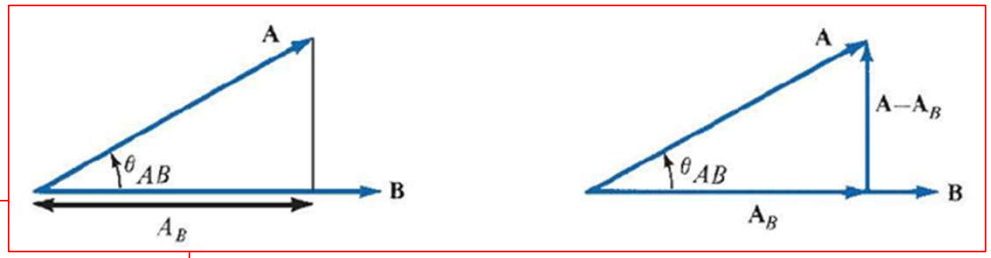
The **dot product** of two vectors **A** and **B**, written as $\mathbf{A} \cdot \mathbf{B}$, is defined geometrically as the product of the magnitudes of **A** and **B** and the cosine of the smaller angle between them when they are drawn tail to tail.

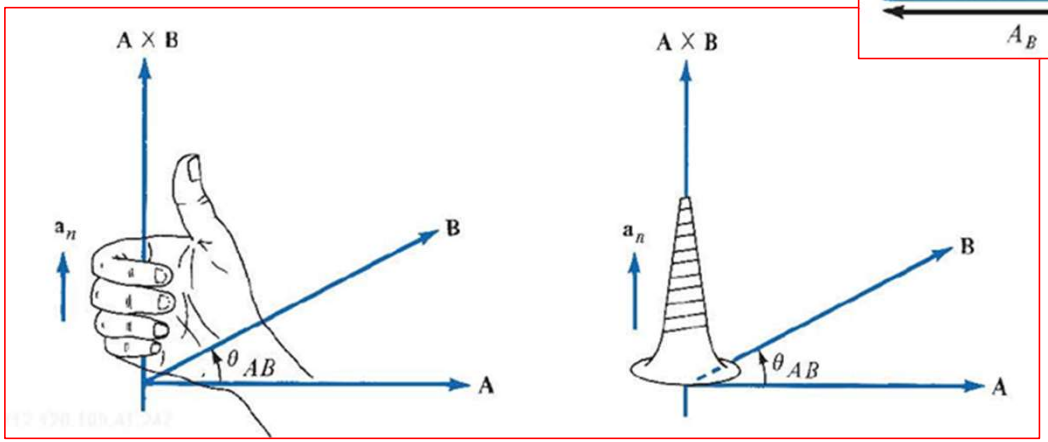
$$\mathbf{A} \cdot \mathbf{B} = AB \cos \theta_{AB}$$

$$\mathbf{a}_x \cdot \mathbf{a}_y = \mathbf{a}_y \cdot \mathbf{a}_z = \mathbf{a}_z \cdot \mathbf{a}_x = 0$$

$$\mathbf{a}_x \cdot \mathbf{a}_x = \mathbf{a}_y \cdot \mathbf{a}_y = \mathbf{a}_z \cdot \mathbf{a}_z = 1$$

$$\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z$$





$$\mathbf{A} \times \mathbf{B} = AB \sin \theta_{AB} \mathbf{a}_n$$

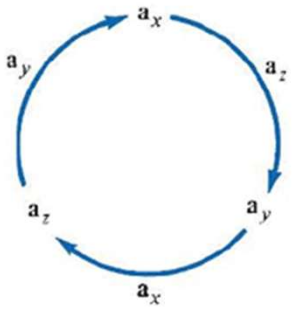
$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$\mathbf{a}_x \times \mathbf{a}_y = \mathbf{a}_z$$

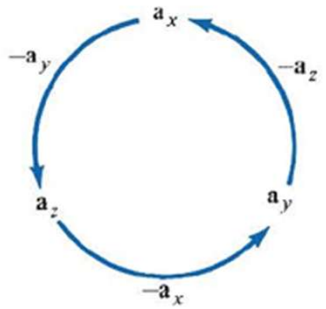
$$\mathbf{a}_y \times \mathbf{a}_z = \mathbf{a}_x$$

$$\mathbf{a}_z \times \mathbf{a}_x = \mathbf{a}_y$$

Cross Product (contd.)



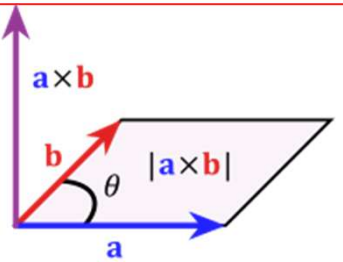
(a)



(b)

$$\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C}) = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}$$

A · (B X C) is the volume of a parallelepiped having **A, B,** and **C** as edges



Given vectors $\mathbf{A} = 3\mathbf{a}_x + 4\mathbf{a}_y + \mathbf{a}_z$ and $\mathbf{B} = 2\mathbf{a}_y - 5\mathbf{a}_z$, find the angle between **A** and **B**.

Solution:

The angle θ_{AB} can be found by using either dot product or cross product.

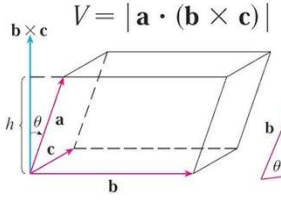
$$\mathbf{A}\cdot\mathbf{B} = (3,4,1)\cdot(0,2,-5)$$
$$= 0 + 8 - 5 = 3$$
$$|\mathbf{A}| = \sqrt{3^2+4^2+1^2} = \sqrt{26}$$
$$|\mathbf{B}| = \sqrt{0^2+2^2+(-5)^2} = \sqrt{29}$$
$$\cos\theta_{AB} = \frac{\mathbf{A}\cdot\mathbf{B}}{|\mathbf{A}||\mathbf{B}|} = \frac{3}{\sqrt{(26)(29)}} = 0.1092$$
$$\theta_{AB} = \cos^{-1} 0.1092 = 83.73°$$

$$\mathbf{A}\times\mathbf{B} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ 3 & 4 & 1 \\ 0 & 2 & -5 \end{vmatrix}$$
$$= (-20-2)\mathbf{a}_x + (0+15)\mathbf{a}_y + (6-0)\mathbf{a}_z$$
$$= (-22, 15, 6)$$
$$|\mathbf{A}\times\mathbf{B}| = \sqrt{(-22)^2+15^2+6^2} = \sqrt{745}$$
$$\sin\theta_{AB} = \frac{|\mathbf{A}\times\mathbf{B}|}{|\mathbf{A}||\mathbf{B}|} = \frac{\sqrt{745}}{\sqrt{(26)(29)}} = 0.994$$
$$\theta_{AB} = \sin^{-1} 0.994 = 83.73°$$

$$\mathbf{a}\times\mathbf{b} = \langle a_2b_3 - a_3b_2,\ a_3b_1 - a_1b_3,\ a_1b_2 - a_2b_1\rangle$$

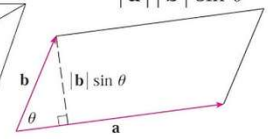
Parallelepiped

$$V = |\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})|$$



Parallelogram

$$A = |\mathbf{a}\times\mathbf{b}| = |\mathbf{a}||\mathbf{b}|\sin\theta$$



If $\mathbf{A} = \mathbf{a}_x + 3\mathbf{a}_z$ and $\mathbf{B} = 5\mathbf{a}_x + 2\mathbf{a}_y - 6\mathbf{a}_z$, find θ_{AB}.

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Answer: 120.6°.

Cross Product (contd.)

The three vertices of a triangle are located at $A(6,-1,2)$, $B(-2,3,-4)$, and $C(-3,1,5)$.
Find:
(*a*) $\mathbf{R}_{AB} \times \mathbf{R}_{AC}$;
(*b*) the area of the triangle;
(*c*) a unit vector perpendicular to the plane in which the triangle is located.

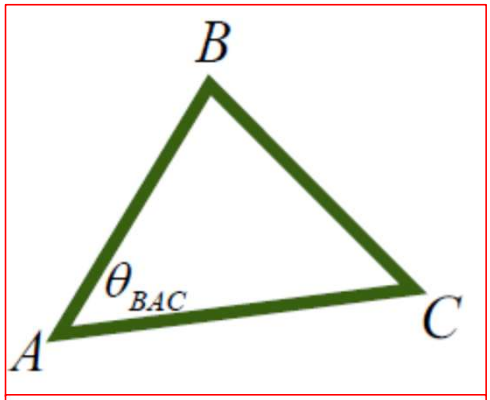
$\mathbf{R_{AB}}$=(-8,4,-6), $\mathbf{R_{AC}}$=(-9,2,3), frame the determinant, $\mathbf{R_{AB}}$x$\mathbf{R_{AC}}$=(12+12, 54+24,-16+36), Find magnitude of cross product

$24\mathbf{a}_x + 78\mathbf{a}_y + 20\mathbf{a}_z$; 42.0; $0.286\mathbf{a}_x + 0.928\mathbf{a}_y + 0.238\mathbf{a}_z$

Dot Product/Cross Product-Problems

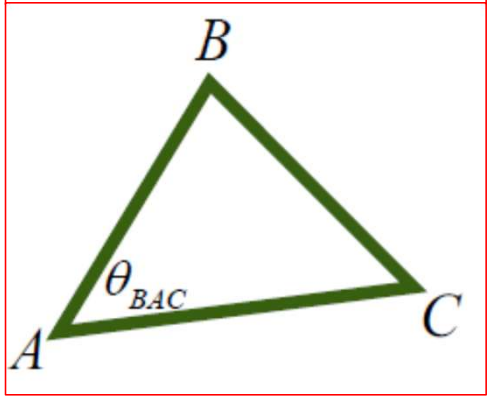
D1.3. The three vertices of a triangle are located at $A(6, -1, 2)$, $B(-2, 3, -4)$, and $C(-3, 1, 5)$. Find: (a) $\mathbf{R}_{AB}$; (b) $\mathbf{R}_{AC}$; (c) the angle θ_{BAC} at vertex A; (d) the (vector) projection of $\mathbf{R}_{AB}$ on $\mathbf{R}_{AC}$.

Ans. $-8\mathbf{a}_x + 4\mathbf{a}_y - 6\mathbf{a}_z$; $-9\mathbf{a}_x + 2\mathbf{a}_y + 3\mathbf{a}_z$; $53.6°$; $-5.94\mathbf{a}_x + 1.319\mathbf{a}_y + 1.979\mathbf{a}_z$

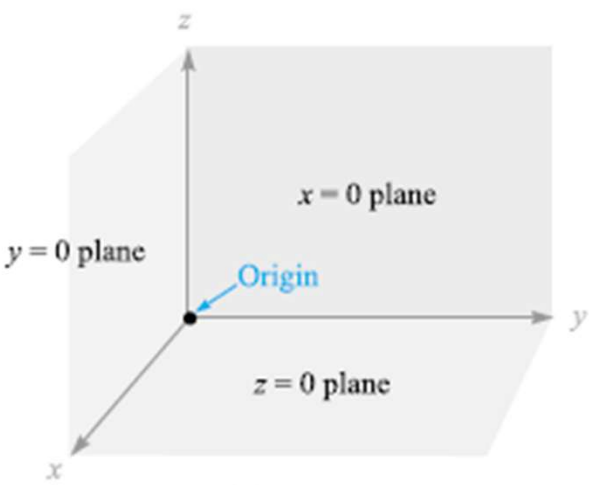


D1.4. The three vertices of a triangle are located at $A(6, -1, 2)$, $B(-2, 3, -4)$, and $C(-3, 1, 5)$. Find: (a) $\mathbf{R}_{AB} \times \mathbf{R}_{AC}$; (b) the area of the triangle; (c) a unit vector perpendicular to the plane in which the triangle is located.

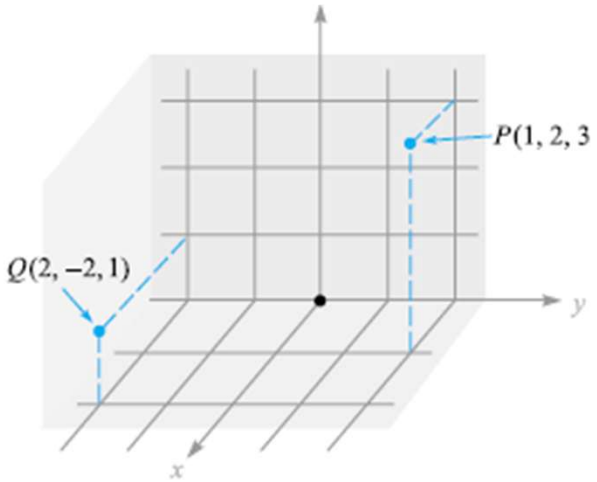
Ans. $24\mathbf{a}_x + 78\mathbf{a}_y + 20\mathbf{a}_z$; 42.0; $0.286\mathbf{a}_x + 0.928\mathbf{a}_y + 0.238\mathbf{a}_z$



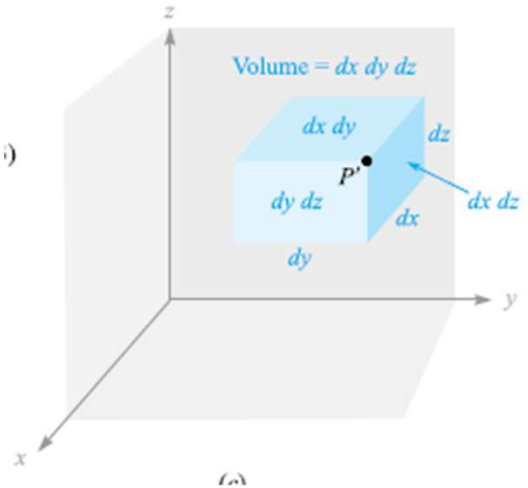
Cartesian or Rectangular Coordinates



(a)



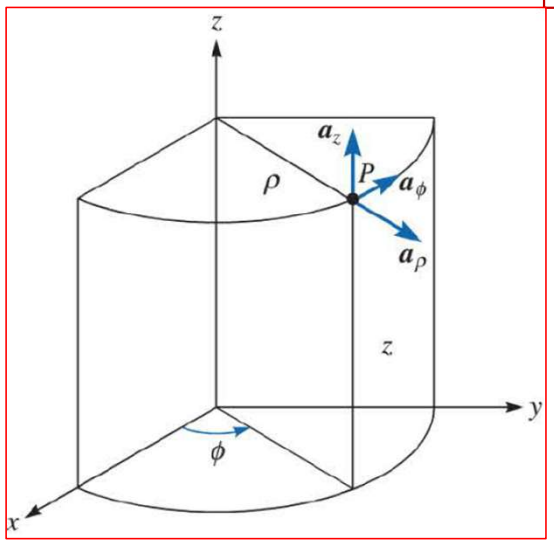
(b)

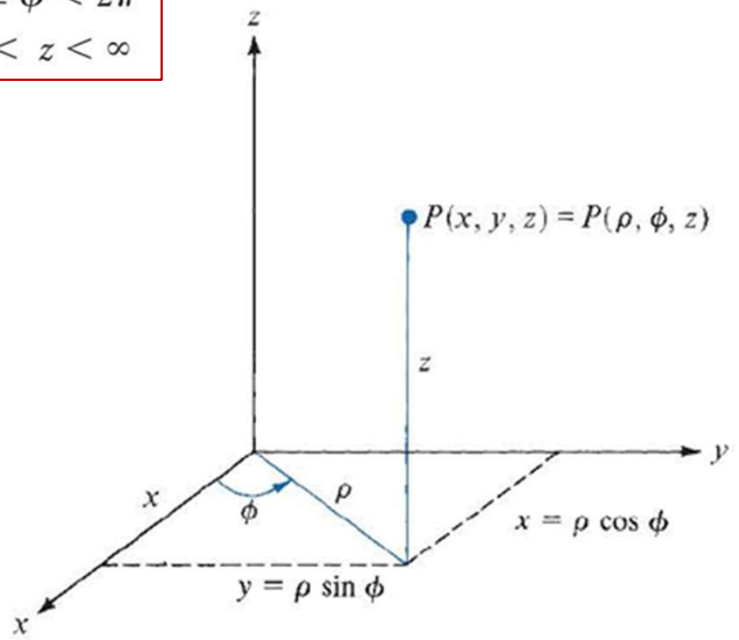


(c)

Cylindrical Co-Ordinate System

$$0 \le \rho < \infty$$
$$0 \le \phi < 2\pi$$
$$-\infty < z < \infty$$





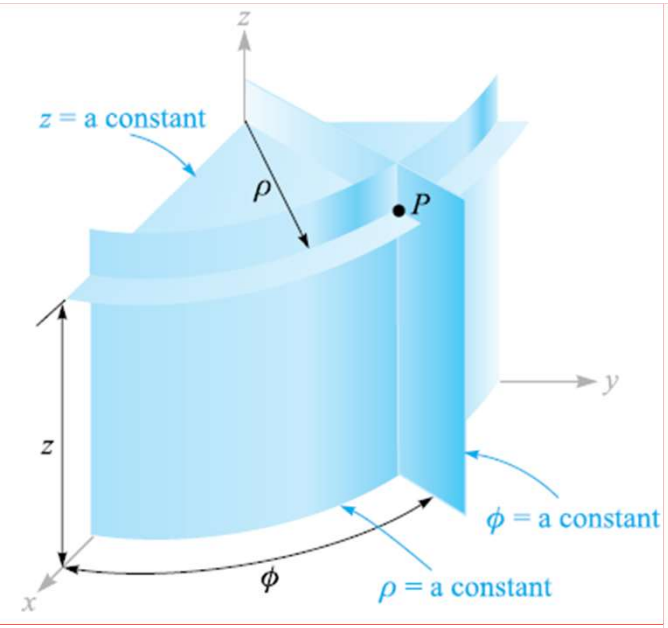
$$\rho = \sqrt{x^2 + y^2}, \quad \phi = \tan^{-1}\frac{y}{x}, \quad z = z$$

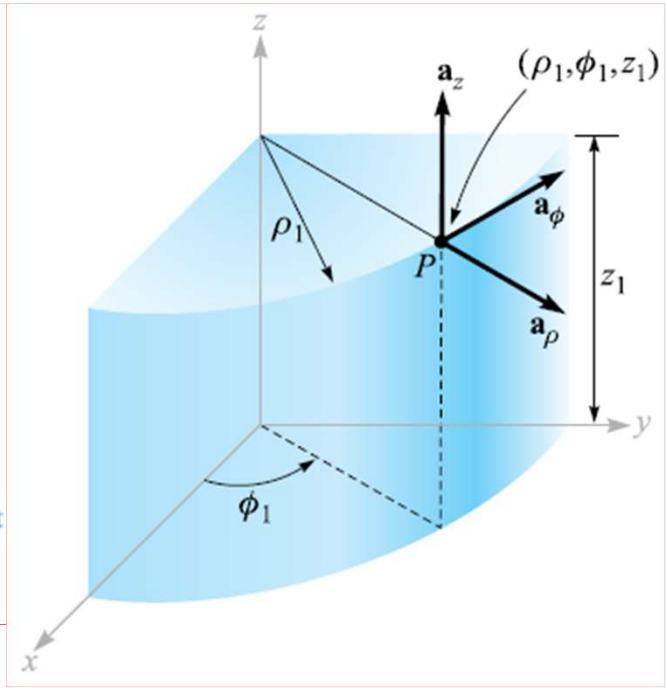
$$x = \rho\cos\phi, \quad y = \rho\sin\phi, \quad z = z$$

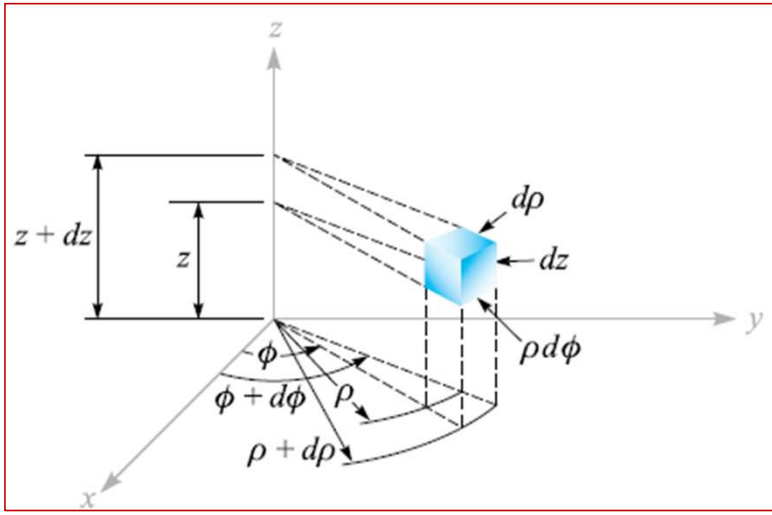
$$(A_\rho, A_\phi, A_z) \quad \text{or} \quad A_\rho\mathbf{a}_\rho + A_\phi\mathbf{a}_\phi + A_z\mathbf{a}_z$$

$$|\mathbf{A}| = (A_\rho^2 + A_\phi^2 + A_z^2)^{1/2}$$

Cylindrical Co-Ordinate System(contd.)







$$\mathbf{a}_\rho \cdot \mathbf{a}_\rho = \mathbf{a}_\phi \cdot \mathbf{a}_\phi = \mathbf{a}_z \cdot \mathbf{a}_z = 1$$

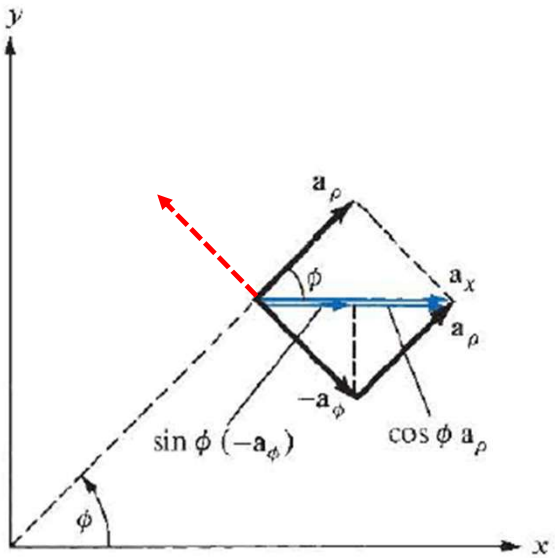
$$\mathbf{a}_\rho \cdot \mathbf{a}_\phi = \mathbf{a}_\phi \cdot \mathbf{a}_z = \mathbf{a}_z \cdot \mathbf{a}_\rho = 0$$

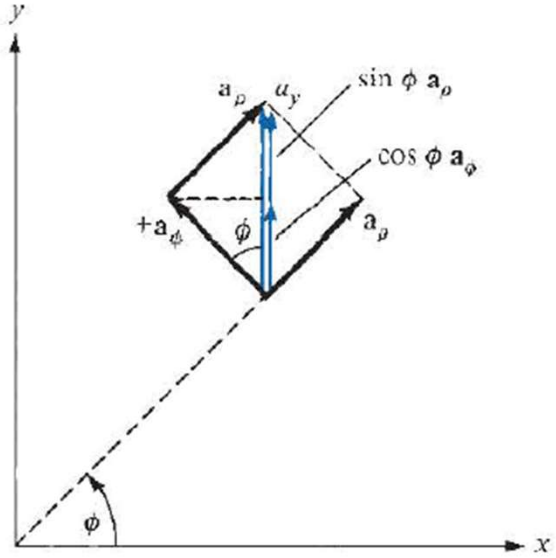
$$\mathbf{a}_\rho \times \mathbf{a}_\phi = \mathbf{a}_z$$

$$\mathbf{a}_\phi \times \mathbf{a}_z = \mathbf{a}_\rho$$

$$\mathbf{a}_z \times \mathbf{a}_\rho = \mathbf{a}_\phi$$

Transformation: Cartesian to Cylindrical Coordinate System





$$\mathbf{a}_x = \cos\phi\,\mathbf{a}_\rho - \sin\phi\,\mathbf{a}_\phi$$
$$\mathbf{a}_y = \sin\phi\,\mathbf{a}_\rho + \cos\phi\,\mathbf{a}_\phi$$
$$\mathbf{a}_z = \mathbf{a}_z$$

$$\cos\phi\,\mathbf{a}_x = \cos^2\phi\,\mathbf{a}_\rho - \cos\phi\sin\phi\,\mathbf{a}_\phi$$
$$\sin\phi\,\mathbf{a}_y = \sin^2\phi\,\mathbf{a}_\rho + \sin\phi\cos\phi\,\mathbf{a}_\phi$$

$$-\sin\phi\,\mathbf{a}_x = -\sin\phi\cos\phi\,\mathbf{a}_\rho + \sin^2\phi\,\mathbf{a}_\phi$$
$$\cos\phi\,\mathbf{a}_y = \sin\phi\cos\phi\,\mathbf{a}_\rho + \cos^2\phi\,\mathbf{a}_\phi$$

$$\mathbf{a}_\rho = \cos\phi\,\mathbf{a}_x + \sin\phi\,\mathbf{a}_y$$
$$\mathbf{a}_\phi = -\sin\phi\,\mathbf{a}_x + \cos\phi\,\mathbf{a}_y$$
$$\mathbf{a}_z = \mathbf{a}_z$$

$\mathbf{a}_x \cdot \mathbf{a}_\rho$

$\mathbf{a}_x \cdot \mathbf{a}_\phi$

$\mathbf{a}_x \cdot \mathbf{a}_z$

$\mathbf{a}_y \cdot \mathbf{a}_\rho$

$\mathbf{a}_y \cdot \mathbf{a}_\phi$

$\mathbf{a}_y \cdot \mathbf{a}_z$

$\mathbf{a}_z \cdot \mathbf{a}_\rho$

$\mathbf{a}_z \cdot \mathbf{a}_\phi$

$\mathbf{a}_z \cdot \mathbf{a}_z$

Transformation: Cartesian to Cylindrical Coordinate System(contd.)

$$\mathbf{a}_x = \cos\phi\,\mathbf{a}_\rho - \sin\phi\,\mathbf{a}_\phi$$
$$\mathbf{a}_y = \sin\phi\,\mathbf{a}_\rho + \cos\phi\,\mathbf{a}_\phi$$
$$\mathbf{a}_z = \mathbf{a}_z$$

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}$$

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \mathbf{a}_x \cdot \mathbf{a}_\rho & \mathbf{a}_x \cdot \mathbf{a}_\phi & \mathbf{a}_x \cdot \mathbf{a}_z \\ \mathbf{a}_y \cdot \mathbf{a}_\rho & \mathbf{a}_y \cdot \mathbf{a}_\phi & \mathbf{a}_y \cdot \mathbf{a}_z \\ \mathbf{a}_z \cdot \mathbf{a}_\rho & \mathbf{a}_z \cdot \mathbf{a}_\phi & \mathbf{a}_z \cdot \mathbf{a}_z \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}$$

Cartesian to Cylindrical	$\mathbf{a}_\rho$	$\mathbf{a}_\phi$	$\mathbf{a}_z$
$\mathbf{a}_x$	$\cos\phi$	$-\sin\phi$	0
$\mathbf{a}_y$	$\sin\phi$	$\cos\phi$	0
$\mathbf{a}_z$	0	0	1

$$\mathbf{a}_\rho = \cos\phi\,\mathbf{a}_x + \sin\phi\,\mathbf{a}_y$$
$$\mathbf{a}_\phi = -\sin\phi\,\mathbf{a}_x + \cos\phi\,\mathbf{a}_y$$
$$\mathbf{a}_z = \mathbf{a}_z$$

$$\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

Transform the vector $\mathbf{B} = y\mathbf{a}_x - x\mathbf{a}_y + z\mathbf{a}_z$ into cylindrical coordinates.

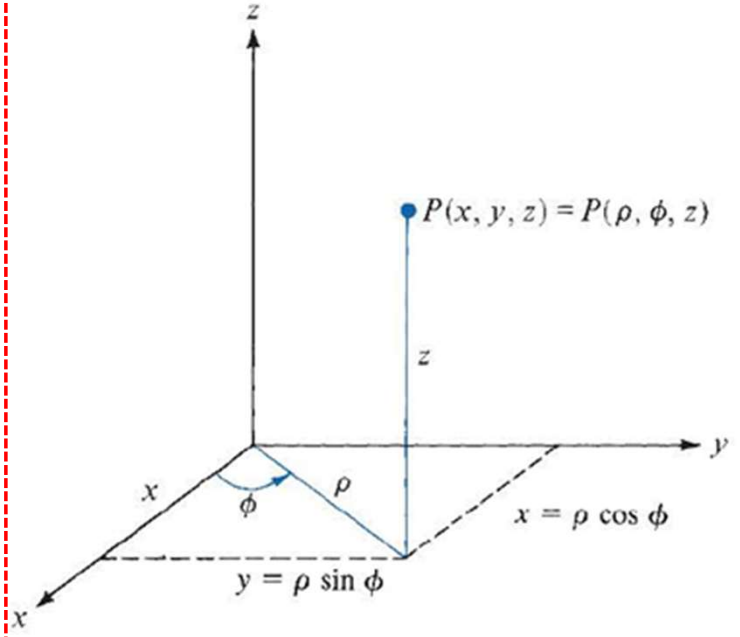
Solution. The new components are

$$\begin{aligned} B_\rho &= \mathbf{B}\cdot\mathbf{a}_\rho = y(\mathbf{a}_x\cdot\mathbf{a}_\rho) - x(\mathbf{a}_y\cdot\mathbf{a}_\rho) \\ &= y\cos\phi - x\sin\phi = \rho\sin\phi\cos\phi - \rho\cos\phi\sin\phi = 0 \\ B_\phi &= \mathbf{B}\cdot\mathbf{a}_\phi = y(\mathbf{a}_x\cdot\mathbf{a}_\phi) - x(\mathbf{a}_y\cdot\mathbf{a}_\phi) \\ &= -y\sin\phi - x\cos\phi = -\rho\sin^2\phi - \rho\cos^2\phi = -\rho \end{aligned}$$

Thus,

$$\mathbf{B} = -\rho\mathbf{a}_\phi + z\mathbf{a}_z$$

Finding unit vector alternative approach



The position vector of any point in cylindrical coordinates is:

$$\vec{r} = x\hat{a}_x + y\hat{a}_y + z\hat{a}_z = \rho\cos\phi\,\hat{a}_x + \rho\sin\phi\,\hat{a}_y + z\hat{a}_z$$

The tangent vectors to the ρ, ϕ, and z curves are given respectively by:

$$\frac{\partial\vec{r}}{\partial\rho} = \cos\phi\,\hat{a}_x + \sin\phi\,\hat{a}_y$$

$$\frac{\partial\vec{r}}{\partial\phi} = -\rho\sin\phi\,\hat{a}_x + \rho\cos\phi\,\hat{a}_y$$

$$\frac{\partial\vec{r}}{\partial z} = \hat{a}_z$$

The unit vectors in these directions are:

$$\vec{a}_\rho = \frac{\partial\vec{r}/\partial\rho}{|\partial\vec{r}/\partial\rho|} = \frac{\cos\phi\,\hat{a}_x + \sin\phi\,\hat{a}_y}{\sqrt{\cos^2\phi + \sin^2\phi}} = \cos\phi\,\hat{a}_x + \sin\phi\,\hat{a}_y$$

$$\vec{a}_\phi = \frac{\partial\vec{r}/\partial\phi}{|\partial\vec{r}/\partial\phi|} = \frac{-\rho\sin\phi\,\hat{a}_x + \rho\cos\phi\,\hat{a}_y}{\sqrt{\rho^2\sin^2\phi + \rho^2\cos^2\phi}} = -\sin\phi\,\hat{a}_x + \cos\phi\,\hat{a}_y$$

$$\vec{a}_z = \frac{\partial\vec{r}}{\partial z} = \hat{a}_z$$

We found a interesting way to find the transformation matrix or the relation between cylindrical and cartesian coordinates.

Problem on Cylindrical Coordinates/Transformation

Convert point P (3, 4, 5) in Cartesian coordinate to cylindrical coordinates?

Transform the vector $\mathbf{B} = y\mathbf{a}_x - x\mathbf{a}_y + z\mathbf{a}_z$ into cylindrical coordinates.

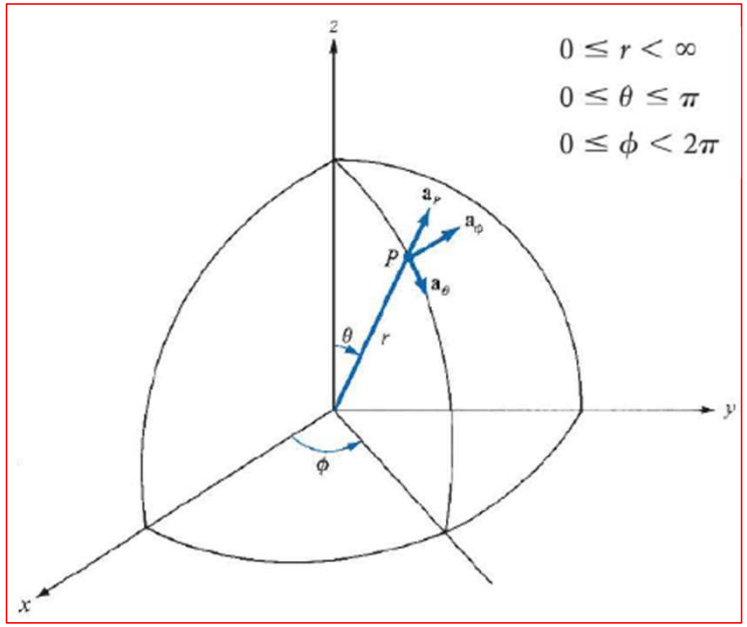
D1.5. (*a*) Give the rectangular coordinates of the point $C(\rho = 4.4, \phi = -115°, z = 2)$. (*b*) Give the cylindrical coordinates of the point $D(x = -3.1, y = 2.6, z = -3)$. (*c*) Specify the distance from C to D.

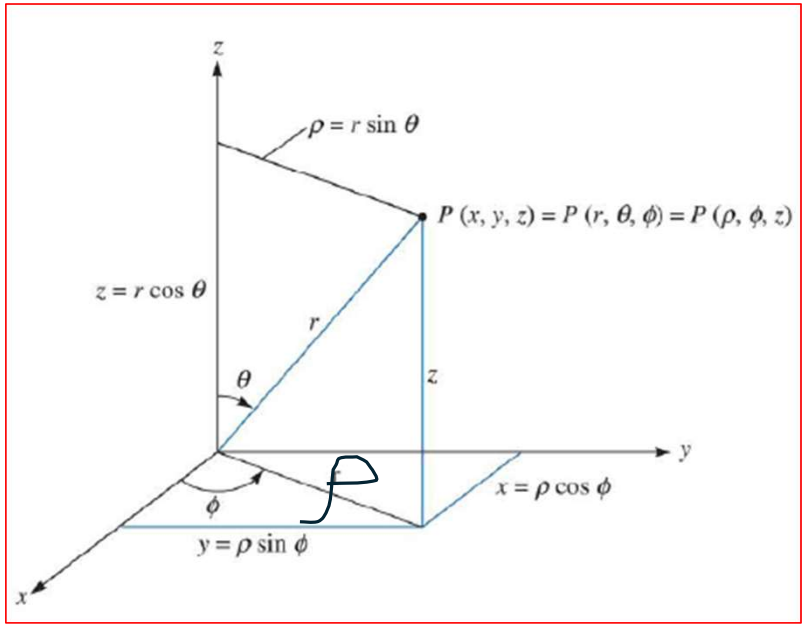
Ans. $C(x = -1.860, y = -3.99, z = 2)$; $D(\rho = 4.05, \phi = 140.0°, z = -3)$; 8.36

D1.6. Transform to cylindrical coordinates: (*a*) $\mathbf{F} = 10\mathbf{a}_x - 8\mathbf{a}_y + 6\mathbf{a}_z$ at point $P(10, -8, 6)$; (*b*) $\mathbf{G} = (2x + y)\mathbf{a}_x - (y - 4x)\mathbf{a}_y$ at point $Q(\rho, \phi, z)$. (*c*) Give the rectangular components of the vector $\mathbf{H} = 20\mathbf{a}_\rho - 10\mathbf{a}_\phi + 3\mathbf{a}_z$ at $P(x = 5, y = 2, z = -1)$.

Ans. $12.81\mathbf{a}_\rho + 6\mathbf{a}_z$; $(2\rho\cos^2\phi - \rho\sin^2\phi + 5\rho\sin\phi\cos\phi)\mathbf{a}_\rho + (4\rho\cos^2\phi - \rho\sin^2\phi - 3\rho\sin\phi\cos\phi)\mathbf{a}_\phi$; $H_x = 22.3$, $H_y = -1.857$, $H_z = 3$

Spherical Polar Coordinates





$$(A_r, A_\theta, A_\phi) \quad \text{or} \quad A_r\mathbf{a}_r + A_\theta\mathbf{a}_\theta + A_\phi\mathbf{a}_\phi$$

$$|\mathbf{A}| = (A_r^2 + A_\theta^2 + A_\phi^2)^{1/2}$$

$$\mathbf{a}_r \cdot \mathbf{a}_r = \mathbf{a}_\theta \cdot \mathbf{a}_\theta = \mathbf{a}_\phi \cdot \mathbf{a}_\phi = 1$$
$$\mathbf{a}_r \cdot \mathbf{a}_\theta = \mathbf{a}_\theta \cdot \mathbf{a}_\phi = \mathbf{a}_\phi \cdot \mathbf{a}_r = 0$$
$$\mathbf{a}_r \times \mathbf{a}_\theta = \mathbf{a}_\phi$$
$$\mathbf{a}_\theta \times \mathbf{a}_\phi = \mathbf{a}_r$$
$$\mathbf{a}_\phi \times \mathbf{a}_r = \mathbf{a}_\theta$$

$$r = \sqrt{x^2 + y^2 + z^2}, \quad \theta = \tan^{-1}\frac{\sqrt{x^2 + y^2}}{z}, \quad \phi = \tan^{-1}\frac{y}{x}$$

$$x = r \sin\theta \cos\phi, \quad y = r \sin\theta \sin\phi, \quad z = r \cos\theta$$

Position Vector in Spherical Coordinates

The position vector of any point in spherical coordinates is given by:

$$\vec{r} = x\hat{a}_x + y\hat{a}_y + z\hat{a}_z = r\sin\theta\cos\phi\,\hat{a}_x + r\sin\theta\sin\phi\,\hat{a}_y + r\cos\theta\,\hat{a}_z$$

1. Partial Derivative w.r.t r

$$\frac{\partial\vec{r}}{\partial r} = \sin\theta\cos\phi\,\hat{a}_x + \sin\theta\sin\phi\,\hat{a}_y + \cos\theta\,\hat{a}_z$$

$$\begin{aligned}
\mathbf{a}_r &= \sin\theta\cos\phi\,\mathbf{a}_x + \sin\theta\sin\phi\,\mathbf{a}_y + \cos\theta\,\mathbf{a}_z \\
\mathbf{a}_\theta &= \cos\theta\cos\phi\,\mathbf{a}_x + \cos\theta\sin\phi\,\mathbf{a}_y - \sin\theta\,\mathbf{a}_z \\
\mathbf{a}_\phi &= -\sin\phi\,\mathbf{a}_x + \cos\phi\,\mathbf{a}_y
\end{aligned}$$

1. $\hat{a}_r$:

$$\hat{a}_r = \frac{\partial\vec{r}/\partial r}{|\partial\vec{r}/\partial r|} = \frac{\sin\theta\cos\phi\,\hat{a}_x + \sin\theta\sin\phi\,\hat{a}_y + \cos\theta\,\hat{a}_z}{\sqrt{\sin^2\theta\cos^2\phi + \sin^2\theta\sin^2\phi + \cos^2\theta}} = \sin\theta\cos\phi\,\hat{a}_x + \sin\theta\sin\phi\,\hat{a}_y + \cos\theta\,\hat{a}_z$$

2. Partial Derivative w.r.t θ

$$\frac{\partial\vec{r}}{\partial\theta} = r\cos\theta\cos\phi\,\hat{a}_x + r\cos\theta\sin\phi\,\hat{a}_y - r\sin\theta\,\hat{a}_z$$

2. $\hat{a}_\theta$:

$$\hat{a}_\theta = \frac{1}{r}\cdot\frac{\partial\vec{r}}{\partial\theta} = \cos\theta\cos\phi\,\hat{a}_x + \cos\theta\sin\phi\,\hat{a}_y - \sin\theta\,\hat{a}_z$$

3. Partial Derivative w.r.t ϕ

$$\frac{\partial\vec{r}}{\partial\phi} = -r\sin\theta\sin\phi\,\hat{a}_x + r\sin\theta\cos\phi\,\hat{a}_y$$

3. $\hat{a}_\phi$:

$$\hat{a}_\phi = \frac{1}{r\sin\theta}\cdot\frac{\partial\vec{r}}{\partial\phi} = -\sin\phi\,\hat{a}_x + \cos\phi\,\hat{a}_y$$

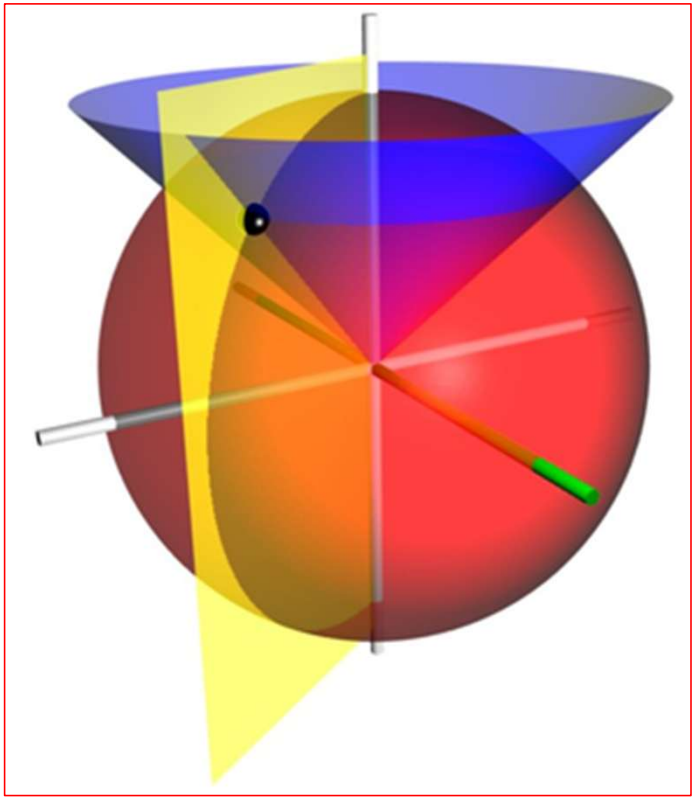
Transformation: Cartesian to Spherical Polar Coordinate System

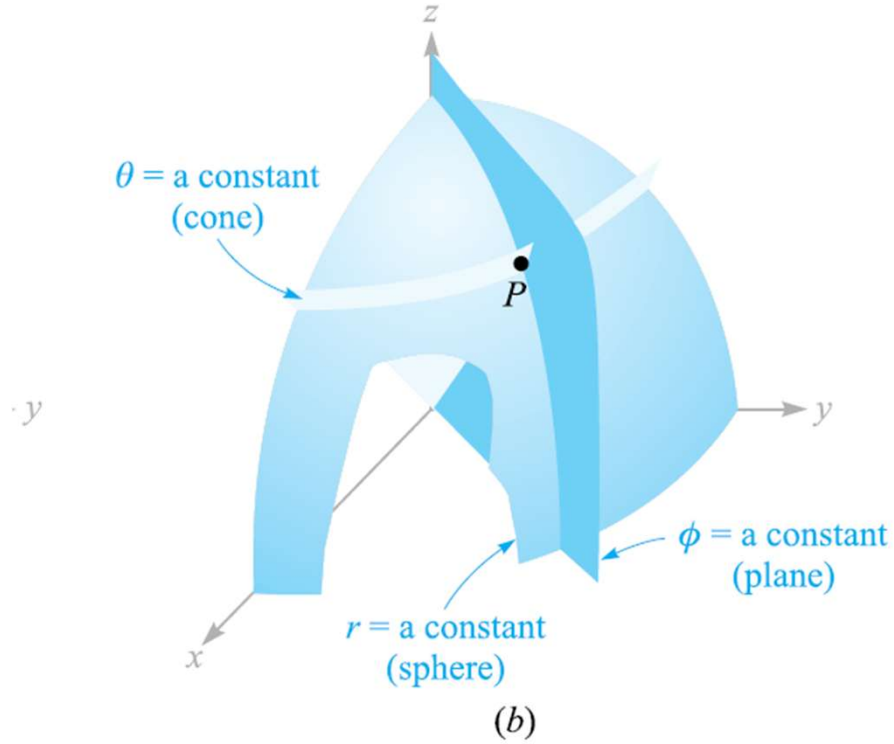
$$\mathbf{a}_x = \sin\theta\cos\phi\,\mathbf{a}_r + \cos\theta\cos\phi\,\mathbf{a}_\theta - \sin\phi\,\mathbf{a}_\phi$$
$$\mathbf{a}_y = \sin\theta\sin\phi\,\mathbf{a}_r + \cos\theta\sin\phi\,\mathbf{a}_\theta + \cos\phi\,\mathbf{a}_\phi$$
$$\mathbf{a}_z = \cos\theta\,\mathbf{a}_r - \sin\theta\,\mathbf{a}_\theta$$

$$\mathbf{a}_r = \sin\theta\cos\phi\,\mathbf{a}_x + \sin\theta\sin\phi\,\mathbf{a}_y + \cos\theta\,\mathbf{a}_z$$
$$\mathbf{a}_\theta = \cos\theta\cos\phi\,\mathbf{a}_x + \cos\theta\sin\phi\,\mathbf{a}_y - \sin\theta\,\mathbf{a}_z$$
$$\mathbf{a}_\phi = -\sin\phi\,\mathbf{a}_x + \cos\phi\,\mathbf{a}_y$$

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \mathbf{a}_r\cdot\mathbf{a}_x & \mathbf{a}_r\cdot\mathbf{a}_y & \mathbf{a}_r\cdot\mathbf{a}_z \\ \mathbf{a}_\theta\cdot\mathbf{a}_x & \mathbf{a}_\theta\cdot\mathbf{a}_y & \mathbf{a}_\theta\cdot\mathbf{a}_z \\ \mathbf{a}_\phi\cdot\mathbf{a}_x & \mathbf{a}_\phi\cdot\mathbf{a}_y & \mathbf{a}_\phi\cdot\mathbf{a}_z \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

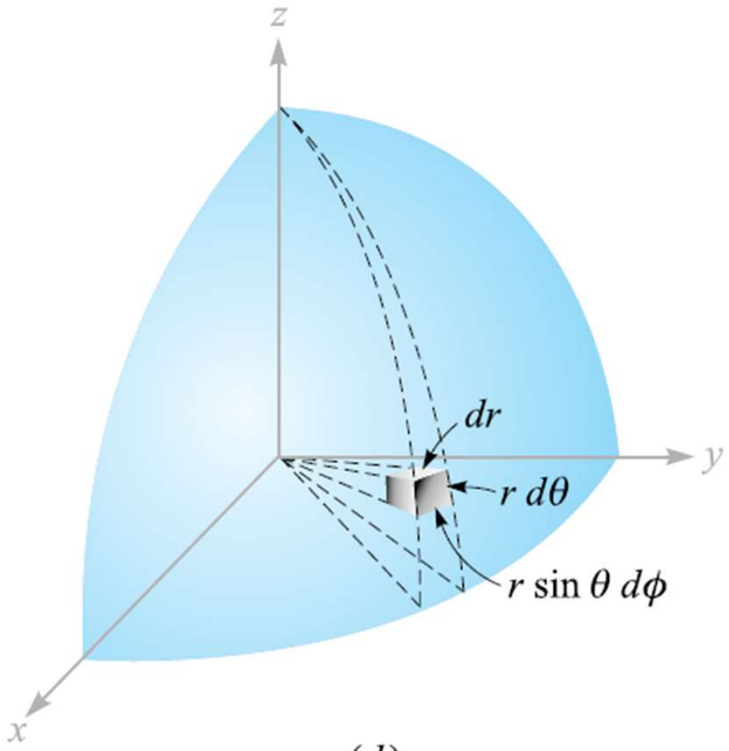
Spherical Polar Coordinate System: Constant Surfaces





https://mathinsight.org/applet/spherical_coordinates_constant_r
https://mathinsight.org/applet/spherical_coordinates_constant_theta
https://mathinsight.org/applet/spherical_coordinates_constant_phi

Spherical Polar Coordinate System: Differential Elements



(*d*)

Problem on Spherical Coordinates/Transformation

D1.7. Given the two points, $C(-3, 2, 1)$ and $D(r = 5, \theta = 20°, \phi = -70°)$, find: (a) the spherical coordinates of C; (b) the rectangular coordinates of D; (c) the distance from C to D.

Ans. $C(r = 3.74, \theta = 74.5°, \phi = 146.3°)$; $D(x = 0.585, y = -1.607, z = 4.70)$; 6.29

D1.8. Transform the following vectors to spherical coordinates at the points given: (a) $10\mathbf{a}_x$ at $P(x = -3, y = 2, z = 4)$; (b) $10\mathbf{a}_y$ at $Q(\rho = 5, \phi = 30°, z = 4)$; (c) $10\mathbf{a}_z$ at $M(r = 4, \theta = 110°, \phi = 120°)$.

Ans. $-5.57\mathbf{a}_r - 6.18\mathbf{a}_\theta - 5.55\mathbf{a}_\phi$; $3.90\mathbf{a}_r + 3.12\mathbf{a}_\theta + 8.66\mathbf{a}_\phi$; $-3.42\mathbf{a}_r - 9.40\mathbf{a}_\theta$

Electrostatics-Coulomb's Law

Coulomb's law states that the force F between two point charges Q_1 and Q_2 is:

1. Along the line joining them
2. Directly proportional to the product Q_1Q_2 of the charges
3. Inversely proportional to the square of the distance R between them.[3]

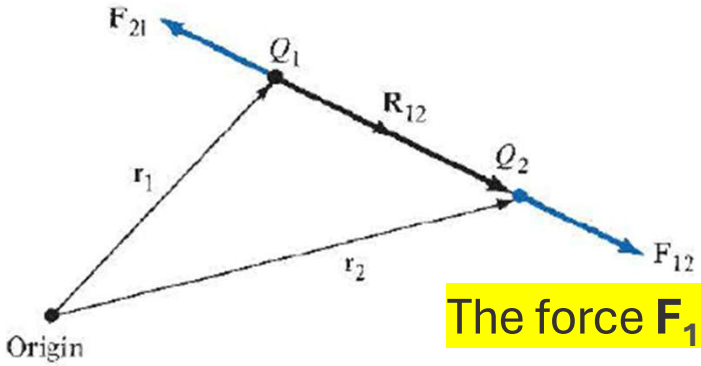
$$F = \frac{k\,Q_1Q_2}{R^2}$$

or

$$\varepsilon_o = 8.854 \times 10^{-12} \simeq \frac{10^{-9}}{36\pi}\ \text{F/m}$$

$$k = \frac{1}{4\pi\varepsilon_o} \simeq 9 \times 10^9\ \text{m/F}$$

The force $\mathbf{F_{21}}$ on Q_1 due to Q_2



$$\mathbf{F}_{12} = \frac{Q_1Q_2}{4\pi\varepsilon_o R^2}\mathbf{a}_{R_{12}}$$

$$\mathbf{R}_{12} = \mathbf{r}_2 - \mathbf{r}_1$$

$$R = |\mathbf{R}_{12}|$$

$$\mathbf{a}_{R_{12}} = \frac{\mathbf{R}_{12}}{R}$$

The force $\mathbf{F_{12}}$ on Q_2 due to Q_1

$$\mathbf{F}_{12} = \frac{Q_1Q_2}{4\pi\varepsilon_o R^3}\mathbf{R}_{12}$$

$$\mathbf{F}_{12} = \frac{Q_1Q_2\,(\mathbf{r}_2 - \mathbf{r}_1)}{4\pi\varepsilon_o|\mathbf{r}_2 - \mathbf{r}_1|^3}$$

$\mathbf{F_{ij}}$:The force $\mathbf{F_{ij}}$ on Q_j due to Q_i

$$\mathbf{F}_{21} = |\mathbf{F}_{12}|\mathbf{a}_{R_{21}} = |\mathbf{F}_{12}|(-\mathbf{a}_{R_{12}})$$

Electrostatics-N Point Charge-Superposition Principle

If we have more than two point charges, we can use the *principle of superposition* to determine the force on a particular charge. The principle states that if there are N charges $Q_1, Q_2, \ldots, Q_N$ located, respectively, at points with position vectors $\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N$, the resultant force $\mathbf{F}$ on a charge Q located at point $\mathbf{r}$ is the vector sum of the forces exerted on Q by each of the charges $Q_1, Q_2, \ldots, Q_N$. Hence,

$$\mathbf{F} = \frac{QQ_1(\mathbf{r} - \mathbf{r}_1)}{4\pi\varepsilon_o|\mathbf{r} - \mathbf{r}_1|^3} + \frac{QQ_2(\mathbf{r} - \mathbf{r}_2)}{4\pi\varepsilon_o|\mathbf{r} - \mathbf{r}_2|^3} + \cdots + \frac{QQ_N(\mathbf{r} - \mathbf{r}_N)}{4\pi\varepsilon_o|\mathbf{r} - \mathbf{r}_N|^3}$$

or

$$\mathbf{F} = \frac{Q}{4\pi\varepsilon_o}\sum_{k=1}^{N}\frac{Q_k(\mathbf{r} - \mathbf{r}_k)}{|\mathbf{r} - \mathbf{r}_k|^3} \qquad (4.8)$$

The **electric field intensity** (or **electric field strength**) $\mathbf{E}$ is the force that a unit positive charge experiences when placed in an electric field.

$$\mathbf{E} = \frac{Q}{4\pi\varepsilon_o R^2}\mathbf{a}_R = \frac{Q(\mathbf{r} - \mathbf{r}')}{4\pi\varepsilon_o|\mathbf{r} - \mathbf{r}'|^3}$$

$$\mathbf{E} = \frac{Q}{4\pi\varepsilon_o r^2}\mathbf{a}_r$$

Electrostatics-Coulomb's Law-Force Calculations

EXAMPLE 2.1

We illustrate the use of the vector form of Coulomb's law by locating a charge of $Q_1 = 3 \times 10^{-4}$ C at $M(1, 2, 3)$ and a charge of $Q_2 = -10^{-4}$ C at $N(2, 0, 5)$ in a vacuum. We desire the force exerted on Q_2 by Q_1.

Solution. We use (3) and (4) to obtain the vector force. The vector $\mathbf{R}_{12}$ is

$$\mathbf{R}_{12} = \mathbf{r}_2 - \mathbf{r}_1 = (2-1)\mathbf{a}_x + (0-2)\mathbf{a}_y + (5-3)\mathbf{a}_z = \mathbf{a}_x - 2\mathbf{a}_y + 2\mathbf{a}_z$$

leading to $|\mathbf{R}_{12}| = 3$, and the unit vector, $\mathbf{a}_{12} = \frac{1}{3}(\mathbf{a}_x - 2\mathbf{a}_y + 2\mathbf{a}_z)$. Thus,

$$\mathbf{F}_2 = \frac{3 \times 10^{-4}(-10^{-4})}{4\pi(1/36\pi)10^{-9} \times 3^2}\left(\frac{\mathbf{a}_x - 2\mathbf{a}_y + 2\mathbf{a}_z}{3}\right)$$

$$= -30\left(\frac{\mathbf{a}_x - 2\mathbf{a}_y + 2\mathbf{a}_z}{3}\right) \text{ N}$$

The magnitude of the force is 30 N, and the direction is specified by the unit vector, which has been left in parentheses to display the magnitude of the force. The force on Q_2 may also be considered as three component forces,

$$\mathbf{F}_2 = -10\mathbf{a}_x + 20\mathbf{a}_y - 20\mathbf{a}_z$$

Electrostatics-Coulomb's Law-Force Calculations

D2.1. A charge $Q_A = -20\,\mu\text{C}$ is located at $A(-6, 4, 7)$, and a charge $Q_B = 50\,\mu\text{C}$ is at $B(5, 8, -2)$ in free space. If distances are given in meters, find: (*a*) $\mathbf{R}_{AB}$; (*b*) R_{AB}. Determine the vector force exerted on Q_A by Q_B if $\epsilon_0 =$ (*c*) $10^{-9}/(36\pi)$ F/m; (*d*) 8.854×10^{-12} F/m.

Ans. $11\mathbf{a}_x + 4\mathbf{a}_y - 9\mathbf{a}_z$ m; 14.76 m; $30.76\mathbf{a}_x + 11.184\mathbf{a}_y - 25.16\mathbf{a}_z$ mN; $30.72\mathbf{a}_x + 11.169\mathbf{a}_y - 25.13\mathbf{a}_z$ mN

Electrostatics-Electrical Field Calculations

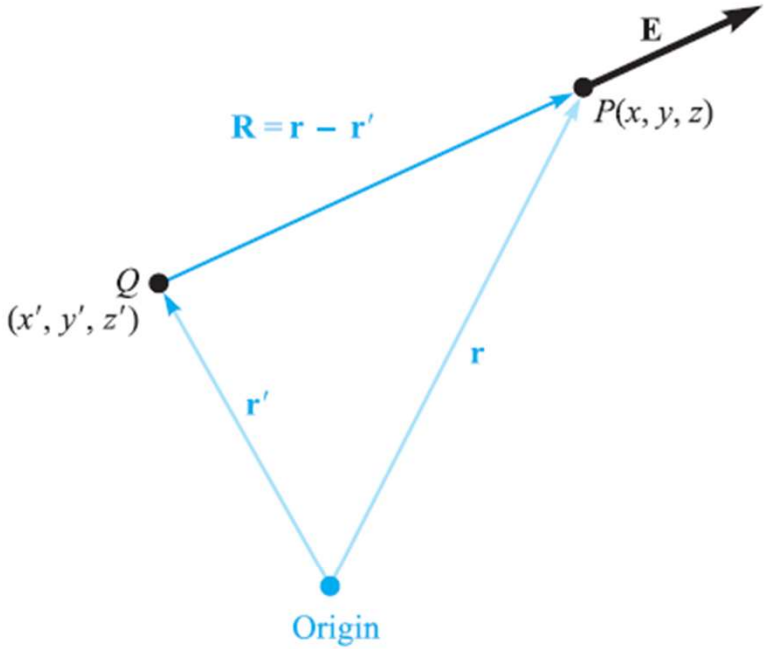
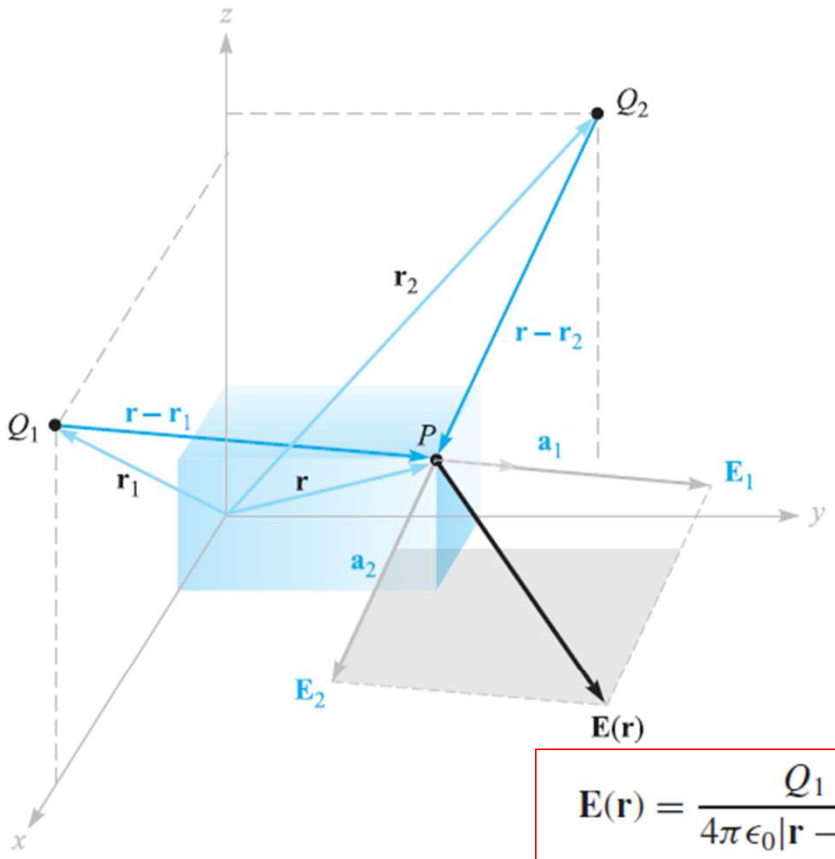


Figure 2.2 The vector r′ locates the point charge Q, the vector r identifies the general point in space $P(x, y, z)$, and the vector R from Q to $P(x, y, z)$ is then $\mathbf{R} = \mathbf{r} - \mathbf{r}'$.



$$\mathbf{E}(\mathbf{r}) = \frac{Q_1}{4\pi\epsilon_0|\mathbf{r}-\mathbf{r}_1|^2}\mathbf{a}_1 + \frac{Q_2}{4\pi\epsilon_0|\mathbf{r}-\mathbf{r}_2|^2}\mathbf{a}_2$$

Figure 2.3 The vector addition of the total electric field intensity at P due to Q_1 and Q_2 is made possible by the linearity of Coulomb's law.

Electrostatics-Electrical Field Calculations

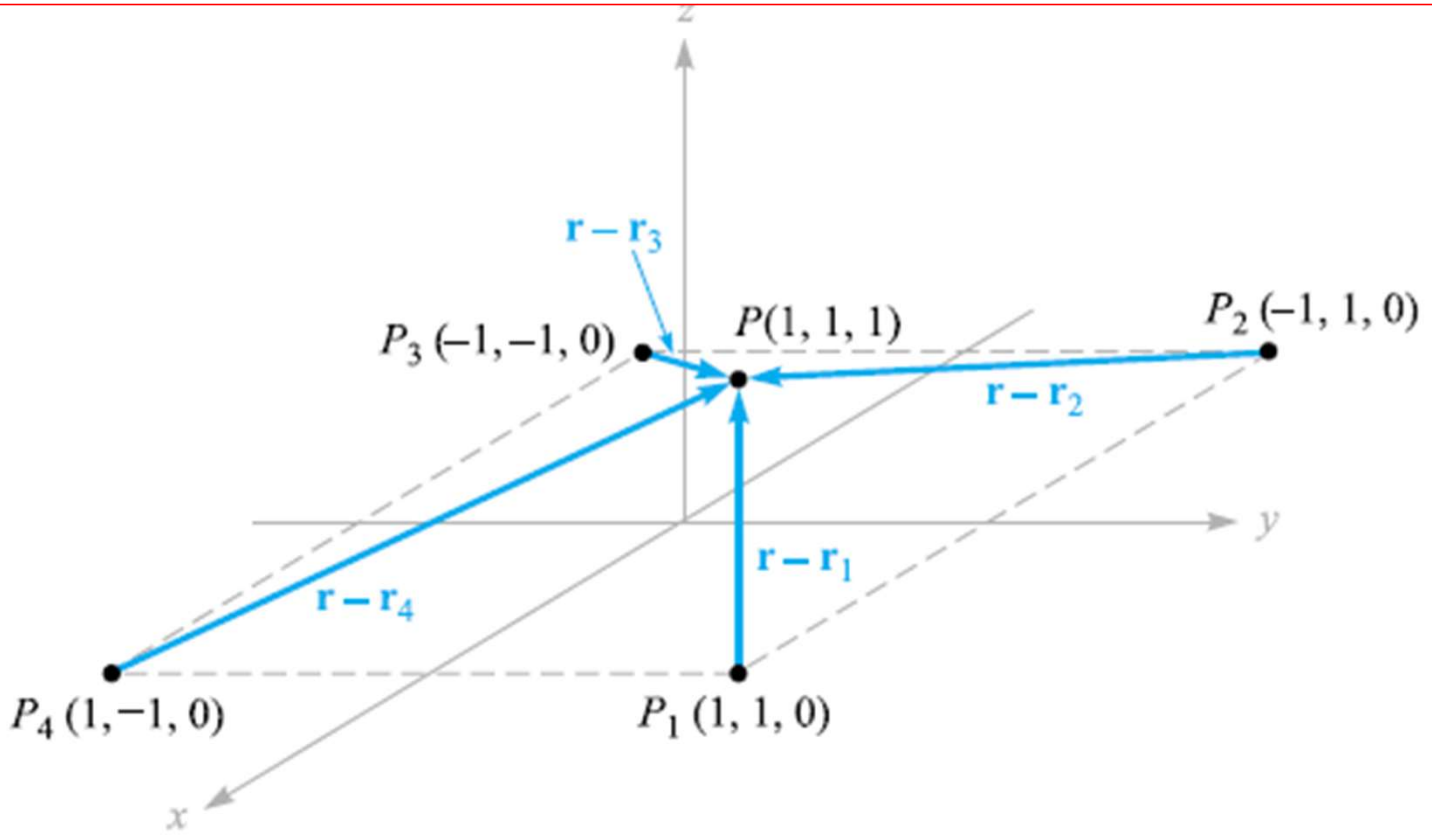


Figure 2.4 A symmetrical distribution of four identical 3-nC point charges produces a field at P, $\mathbf{E} = 6.82\mathbf{a}_x + 6.82\mathbf{a}_y + 32.8\mathbf{a}_z$ V/m.

Electrostatics-Electrical Field Calculations

EXAMPLE 2.2

In order to illustrate the application of (11), we find $\mathbf{E}$ at $P(1, 1, 1)$ caused by four identical 3-nC (nanocoulomb) charges located at $P_1(1, 1, 0)$, $P_2(-1, 1, 0)$, $P_3(-1, -1, 0)$, and $P_4(1, -1, 0)$, as shown in Figure 2.4.

Solution. We find that $\mathbf{r} = \mathbf{a}_x + \mathbf{a}_y + \mathbf{a}_z$, $\mathbf{r}_1 = \mathbf{a}_x + \mathbf{a}_y$, and thus $\mathbf{r} - \mathbf{r}_1 = \mathbf{a}_z$. The magnitudes are: $|\mathbf{r} - \mathbf{r}_1| = 1$, $|\mathbf{r} - \mathbf{r}_2| = \sqrt{5}$, $|\mathbf{r} - \mathbf{r}_3| = 3$, and $|\mathbf{r} - \mathbf{r}_4| = \sqrt{5}$. Because $Q/4\pi\epsilon_0 = 3 \times 10^{-9}/(4\pi \times 8.854 \times 10^{-12}) = 26.96\ \text{V} \cdot \text{m}$, we may now use (11) to obtain

$$\mathbf{E} = 26.96\left[\frac{\mathbf{a}_z}{1}\frac{1}{1^2} + \frac{2\mathbf{a}_x + \mathbf{a}_z}{\sqrt{5}}\frac{1}{(\sqrt{5})^2} + \frac{2\mathbf{a}_x + 2\mathbf{a}_y + \mathbf{a}_z}{3}\frac{1}{3^2} + \frac{2\mathbf{a}_y + \mathbf{a}_z}{\sqrt{5}}\frac{1}{(\sqrt{5})^2}\right]$$

or

$$\mathbf{E} = 6.82\mathbf{a}_x + 6.82\mathbf{a}_y + 32.8\mathbf{a}_z\ \text{V/m}$$

Electrostatics-Electrical Field Calculations

■ Example
A charge Q_1 of 2 μC is located at at $P_1(0,0,0)$ and a second charge of 3 μC is at $P_2(-1,2,3)$. Find E at $M(3,-4,2)$.

$= 623.7\mathbf{a}_x - 879.92\mathbf{a}_y + 160.17\mathbf{a}_z$ V/m

D2.2. A charge of $-0.3\,\mu$C is located at $A(25, -30, 15)$ (in cm), and a second charge of $0.5\,\mu$C is at $B(-10, 8, 12)$ cm. Find E at: (*a*) the origin; (*b*) $P(15, 20, 50)$ cm.

Ans. $92.3\mathbf{a}_x - 77.6\mathbf{a}_y - 94.2\mathbf{a}_z$ kV/m; $11.9\mathbf{a}_x - 0.519\mathbf{a}_y + 12.4\mathbf{a}_z$ kV/m

Line-Surface-Volume Charge Density-Total Charge Calculation

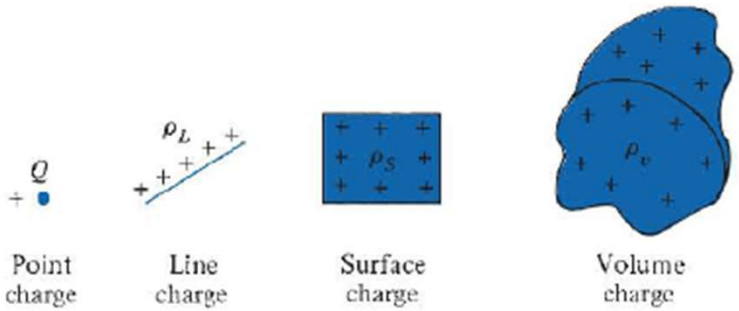


FIGURE 4.5 Various charge distributions and charge elements.

$$dQ = \rho_L\, dl \rightarrow Q = \int_L \rho_L\, dl \quad \text{(line charge)}$$

$$dQ = \rho_S\, dS \rightarrow Q = \int_S \rho_S\, dS \quad \text{(surface charge)}$$

$$dQ = \rho_v\, dv \rightarrow Q = \int_v \rho_v\, dv \quad \text{(volume charge)}$$

$$\mathbf{E} = \int_L \frac{\rho_L\, dl}{4\pi\varepsilon_o R^2}\, \mathbf{a}_R \quad \text{(line charge)}$$

$$\mathbf{E} = \int_S \frac{\rho_S\, dS}{4\pi\varepsilon_o R^2}\, \mathbf{a}_R \quad \text{(surface charge)}$$

$$\mathbf{E} = \int_v \frac{\rho_v\, dv}{4\pi\varepsilon_o R^2}\, \mathbf{a}_R \quad \text{(volume charge)}$$

Electrostatics-Volume Charge Density-Total Charge Calculation

We denote volume charge density by ρ_v, having the units of coulombs per cubic meter (C/m^3).

The small amount of charge ΔQ in a small volume Δv is

$$\Delta Q = \rho_v \Delta v \tag{12}$$

and we may define ρ_v mathematically by using a limiting process on (12),

$$\rho_v = \lim_{\Delta v \to 0} \frac{\Delta Q}{\Delta v} \tag{13}$$

The total charge within some finite volume is obtained by integrating throughout that volume,

$$Q = \int_{\text{vol}} \rho_v dv \tag{14}$$

Only one integral sign is customarily indicated, but the differential dv signifies integration throughout a volume, and hence a triple integration.

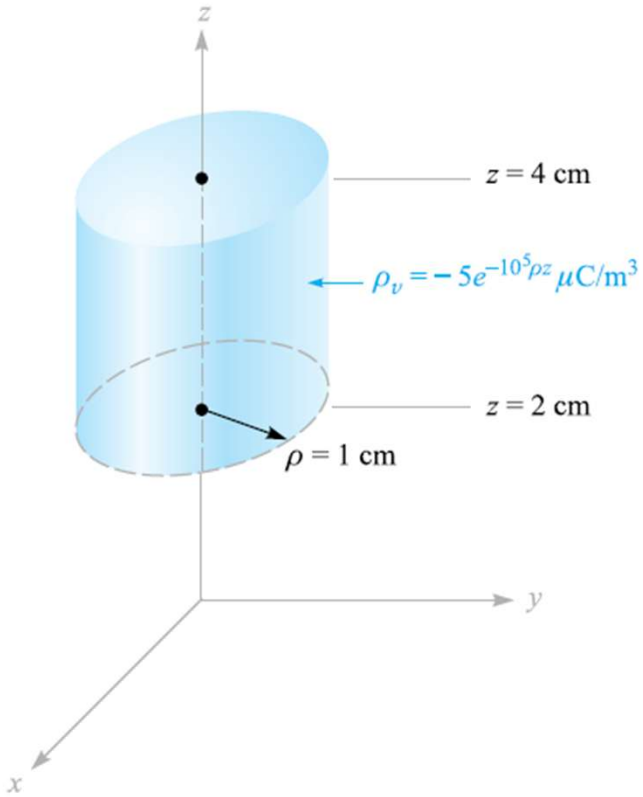


Figure 2.5 The total charge contained within the right circular cylinder may be obtained by evaluating $Q = \int_{\text{vol}} \rho_v dv$.

Find the total charge inside the volume indicated by $\rho_v = 4xyz^2$, $0 \le \rho \le 2$, $0 \le \Phi \le \pi/2$, $0 \le z \le 3$. All values are in SI units.

Electrostatics-Volume Charge Density-Total Charge Calculation

D2.4. Calculate the total charge within each of the indicated volumes: (*a*) $0.1 \leq |x|, |y|, |z| \leq 0.2$: $\rho_v = \dfrac{1}{x^3 y^3 z^3}$; (*b*) $0 \leq \rho \leq 0.1, 0 \leq \phi \leq \pi, 2 \leq z \leq 4$; $\rho_v = \rho^2 z^2 \sin 0.6\phi$; (*c*) universe: $\rho_v = e^{-2r}/r^2$.

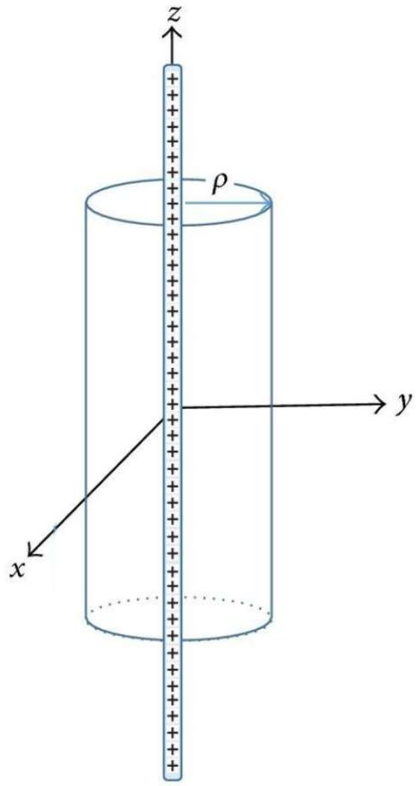
Ans. 0; 1.018 mC; 6.28 C

Field Due to Line Charge Distribution

- Now we consider a filament like distribution of volume charge density. It is convenient to treat the charge as a line charge of density ρ_L C/m.

- We desire the electric field intensity E at any point resulting from a uniform line charge density ρ_L

- The field falls off inversely with the distance to the charged line, as compared with the point charge, where the field decreased with the square of the distance.



$$\mathbf{E} = \frac{\rho_L}{2\pi\varepsilon_0\rho}\mathbf{a}_\rho$$

(Proof: Not included for exam)

Field Due to Line Charge Distribution

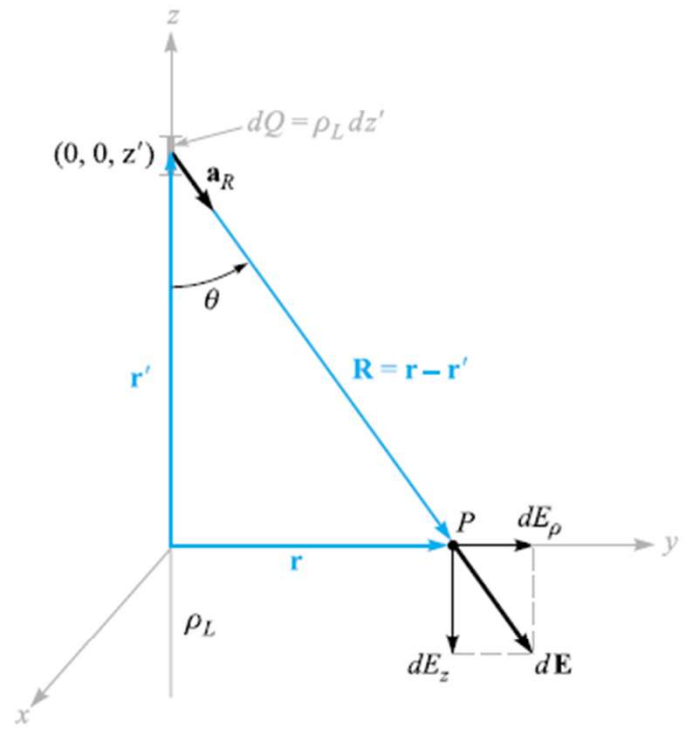


Figure 2.6 The contribution $d\mathbf{E} = dE_\rho \mathbf{a}_\rho + dE_z \mathbf{a}_z$ to the electric field intensity produced by an element of charge $dQ = \rho_L dz'$ located a distance z' from the origin. The linear charge density is uniform and extends along the entire z axis.

$$d\mathbf{E} = \frac{\rho_L dz'(\mathbf{r} - \mathbf{r}')}{4\pi\epsilon_0|\mathbf{r} - \mathbf{r}'|^3}$$

$$\mathbf{r} = y\mathbf{a}_y = \rho\mathbf{a}_\rho$$

$$\mathbf{r}' = z'\mathbf{a}_z$$

$$\mathbf{r} - \mathbf{r}' = \rho\mathbf{a}_\rho - z'\mathbf{a}_z$$

$$d\mathbf{E} = \frac{\rho_L dz'(\rho\mathbf{a}_\rho - z'\mathbf{a}_z)}{4\pi\epsilon_0(\rho^2 + z'^2)^{3/2}}$$

$$E_\rho = \int_{-\infty}^{\infty} \frac{\rho_L \rho dz'}{4\pi\epsilon_0(\rho^2 + z'^2)^{3/2}}$$

Coordinate	Does Field Vary?	Component Present?	Reason
ϕ	No	$E_\phi = 0$	Azimuthal symmetry
z	No	$E_z = 0$	Axial symmetry and cancellation
ρ	Yes	$E_\rho \neq 0$	Field strength changes with distance

$$\mathbf{E} = \frac{\rho_L}{2\pi\varepsilon_0\rho}\mathbf{a}_\rho$$

Field Due to Line Charge Distribution

Consider an infinite line charge parallel to the z axis at $x = 6$, $y = 8$. Find **E** at the general field point $P(x, y, z)$.

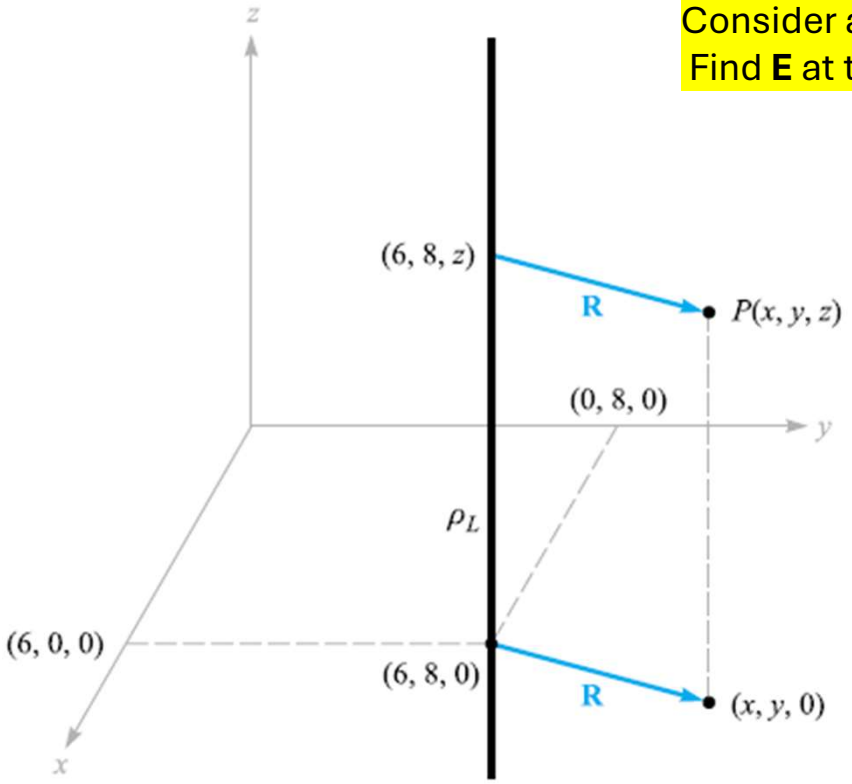


Figure 2.7 A point $P(x, y, z)$ is identified near an infinite uniform line charge located at $x = 6$, $y = 8$.

$$\mathbf{E} = \frac{\rho_L}{2\pi\epsilon_0\sqrt{(x-6)^2+(y-8)^2}}\mathbf{a}_R$$

$$\mathbf{a}_R = \frac{\mathbf{R}}{|\mathbf{R}|} = \frac{(x-6)\mathbf{a}_x + (y-8)\mathbf{a}_y}{\sqrt{(x-6)^2+(y-8)^2}}$$

$$\mathbf{E} = \frac{\rho_L}{2\pi\epsilon_0}\frac{(x-6)\mathbf{a}_x + (y-8)\mathbf{a}_y}{(x-6)^2+(y-8)^2}$$

Note that the field is not a function of z.

Field Due to Line Charge Distribution-Problems

D2.5. Infinite uniform line charges of 5 nC/m lie along the (positive and negative) x and y axes in free space. Find E at: (a) $P_A(0, 0, 4)$; (b) $P_B(0, 3, 4)$.

Ans. $45\mathbf{a}_z$ V/m; $10.8\mathbf{a}_y + 36.9\mathbf{a}_z$ V/m

Problem

On the line described by

$$x = 2\,\text{m}, \quad y = -4\,\text{m}$$

there is a uniform charge distribution of density

$$\rho_\ell = 20\,\text{nC/m}.$$

Determine the electric field $\vec{E}$ at the point

$$(-2,\ -1,\ 4)\,\text{m}.$$

Field Due to Sheet of Charge (Surface Charge)

- Another basic charge configuration is the infinite sheet of charge having a uniform density of ρ_S C/m^2.
- The charge-distribution family is now complete: point (Q), line (ρ_L), surface (ρ_S), and volume (ρ_v).
- Let us examine a sheet of charge, which is placed in the yz plane.
- The plane can be seen to be assembled from an infinite number of line charge, extending along the z axis, from –∞ to +∞.

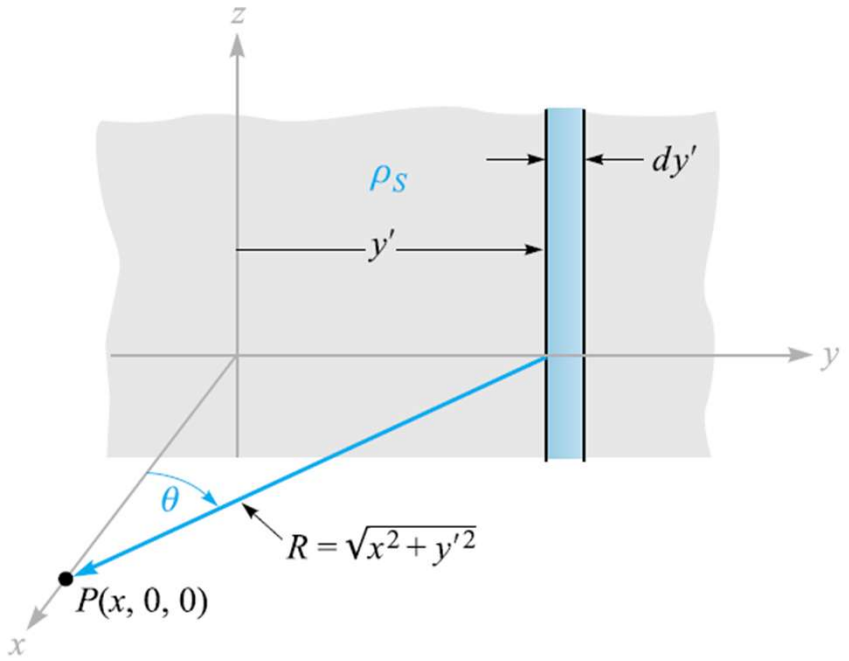


Figure 2.8 An infinite sheet of charge in the *yz* plane, a general point *P* on the *x* axis, and the differential-width line charge used as the element in determining the field at *P* by $d\mathbf{E} = \rho_S dy' \mathbf{a}_R/(2\pi\varepsilon_0 R)$.

Field Due to Sheet of Charge (Surface Charge)

- A unit vector a_N, which is normal to the sheet and directed away from it.
- The field of a sheet of charge is constant in magnitude and direction.
- It is not a function of distance.
- Fact: The electric field is always directed away from the positive charge, into the negative charge.
- If the point *P* were chosen on the negative *x* axis, then what will the field?

$$\mathbf{E} = \frac{\rho_s}{2\varepsilon_0}\mathbf{a}_N$$

$$E_x = -\frac{\rho_S}{2\epsilon_0}$$

Field Due to Sheet of Charge-Different Problem

If a second infinite sheet of charge, having a *negative* charge density $-\rho_S$, is located in the plane $x = a$, we may find the total field by adding the contribution of each sheet. In the region $x > a$,

$$\mathbf{E}_+ = \frac{\rho_S}{2\epsilon_0}\mathbf{a}_x \qquad \mathbf{E}_- = -\frac{\rho_S}{2\epsilon_0}\mathbf{a}_x \qquad \mathbf{E} = \mathbf{E}_+ + \mathbf{E}_- = 0$$

and for $x < 0$,

$$\mathbf{E}_+ = -\frac{\rho_S}{2\epsilon_0}\mathbf{a}_x \qquad \mathbf{E}_- = \frac{\rho_S}{2\epsilon_0}\mathbf{a}_x \qquad \mathbf{E} = \mathbf{E}_+ + \mathbf{E}_- = 0$$

and when $0 < x < a$,

$$\mathbf{E}_+ = \frac{\rho_S}{2\epsilon_0}\mathbf{a}_x \qquad \mathbf{E}_- = \frac{\rho_S}{2\epsilon_0}\mathbf{a}_x$$

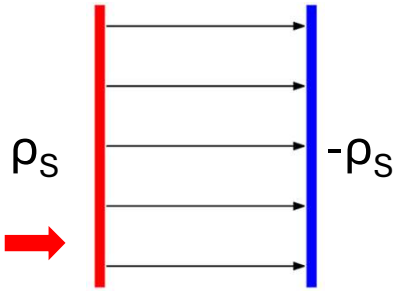
and

$$\mathbf{E} = \mathbf{E}_+ + \mathbf{E}_- = \frac{\rho_S}{\epsilon_0}\mathbf{a}_x \qquad (18)$$

Physical Picture

- Think of each plate separately:
  - A single infinite charged plate makes a **constant** field everywhere.
  - Two such plates with **opposite charges** → inside fields add, outside fields cancel.
- For finite plates, the cancellation is incomplete outside → small fringing fields exist due to leaking of charge from edge.

Parallel Plate Capacitor



Field Due to Sheet of Charge (Surface Charge)- Problems

2.25 Find **E** at the origin if the following charge distributions are present in free space: point charge, 12 nC, at $P(2, 0, 6)$; uniform line charge density, 3 nC/m, at $x = -2$, $y = 3$; uniform surface charge density, 0.2 nC/m^2, at $x = 2$.

Field for Point Charge | Field for Line Charge | Field for Surface Charge

$$\mathbf{E} = \left[\frac{(12 \times 10^{-9})}{4\pi\epsilon_0}\frac{(-2\mathbf{a}_x - 6\mathbf{a}_z)}{(4+36)^{1.5}}\right] + \left[\frac{(3 \times 10^{-9})}{2\pi\epsilon_0}\frac{(2\mathbf{a}_x - 3\mathbf{a}_y)}{(4+9)}\right] - \left[\frac{(0.2 \times 10^{-9})\mathbf{a}_x}{2\epsilon_0}\right]$$

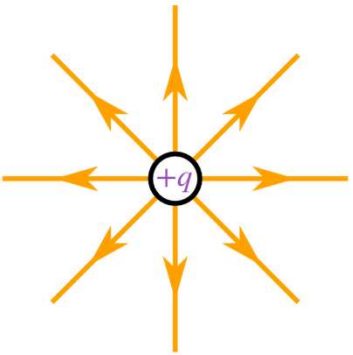
$$= -3.9\mathbf{a}_x - 12.4\mathbf{a}_y - 2.5\mathbf{a}_z \text{ V/m}$$

D2.6. Three infinite uniform sheets of charge are located in free space as follows: 3 nC/m^2 at $z = -4$, 6 nC/m^2 at $z = 1$, and -8 nC/m^2 at $z = 4$. Find **E** at the point: (*a*) $P_A(2, 5, -5)$; (*b*) $P_B(4, 2, -3)$; (*c*) $P_C(-1, -5, 2)$; (*d*) $P_D(-2, 4, 5)$.

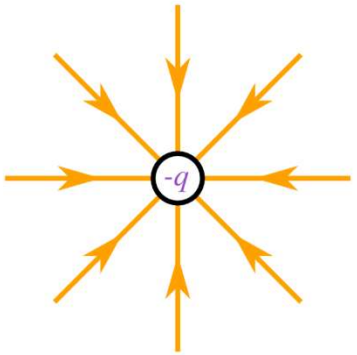
Ans. $-56.5\mathbf{a}_z$; $283\mathbf{a}_z$; $961\mathbf{a}_z$; $56.5\mathbf{a}_z$ all V/m

Electric Field Line and Flux

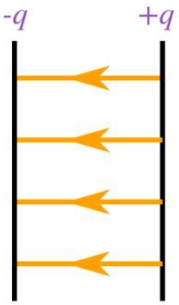
E field around a positive charge

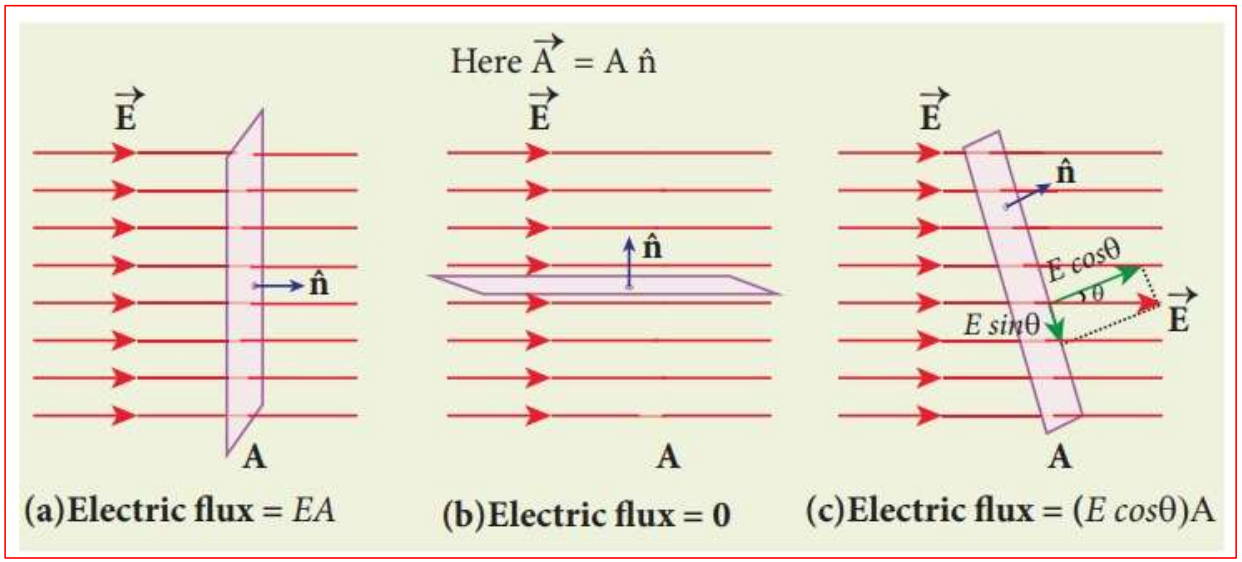


E field around a negative charge



E field inside a parallel plate capacitor





Electric Flux Density

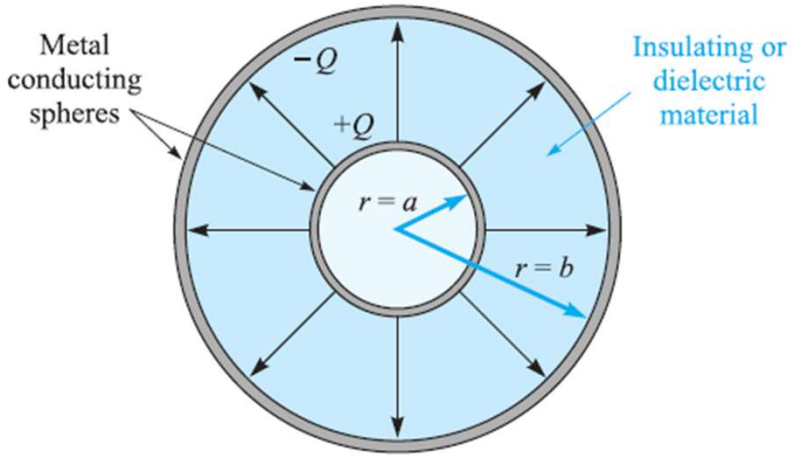


Figure 3.1 The electric flux in the region between a pair of charged concentric spheres. The direction and magnitude of D are not functions of the dielectric between the spheres.

- At the surface of the inner sphere, ψ coulombs of electric flux are produced by the given charge Q coulombs, and distributed uniformly over a surface having an area of 4πa2 m2.
- The density of the flux at this surface is ψ/ 4πa2 or Q/ 4πa2 C/m2
- The electric flux density is in the radial direction

$$\mathbf{D}\Big|_{r=a} = \frac{Q}{4\pi a^2}\mathbf{a}_r \qquad \text{(inner sphere)}$$

$$\mathbf{D}\Big|_{r=b} = \frac{Q}{4\pi b^2}\mathbf{a}_r \qquad \text{(outer sphere)}$$

- At a distance *r*, where *a*≤ *r*≤ *b*,

$$\mathbf{D} = \frac{Q}{4\pi r^2}\mathbf{a}_r$$

Electric Flux Density

- If we make the inner sphere smaller and smaller, it becomes a point charge while still retaining a charge of Q. The electric flux density at a point **r** meters away is still given by:

$$\mathbf{D} = \frac{Q}{4\pi r^2}\mathbf{a}_r$$

- The radial electric field intensity of a point charge in free space,

$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0 r^2}\mathbf{a}_r$$

- In free space, therefore

$$\mathbf{D} = \epsilon_0\mathbf{E} \quad \text{(free space only)}$$

For a general volume charge distribution in free space

$$\mathbf{E} = \int_{\text{vol}} \frac{\rho_v dv}{4\pi\epsilon_0 R^2}\mathbf{a}_R \quad \text{(free space only)}$$

$$\mathbf{D} = \int_{\text{vol}} \frac{\rho_v dv}{4\pi R^2}\mathbf{a}_R$$

Electric Flux Density-Problem

■ Example

Find the electric flux density at a point having a distance 3 m from a uniform line charge of 8 nC/m lying along the *z* axis in free space.

$$\mathbf{E} = \frac{\rho_L}{2\pi\varepsilon_0\rho}\mathbf{a}_\rho \Rightarrow \mathbf{D} = \frac{\rho_L}{2\pi\rho}\mathbf{a}_\rho = \frac{8\times10^{-9}}{2\pi\rho}\mathbf{a}_\rho = \frac{1.273\times10^{-9}}{\rho}\mathbf{a}_\rho \ \mathrm{C/m^2}$$

For the value ρ = 3 m,

$$\mathbf{D} = \frac{1.273\times10^{-9}}{3} = 4.244\times10^{-10}\mathbf{a}_\rho \ \mathrm{C/m^2} = \underline{\underline{0.424\mathbf{a}_\rho \ \mathrm{nC/m^2}}}$$

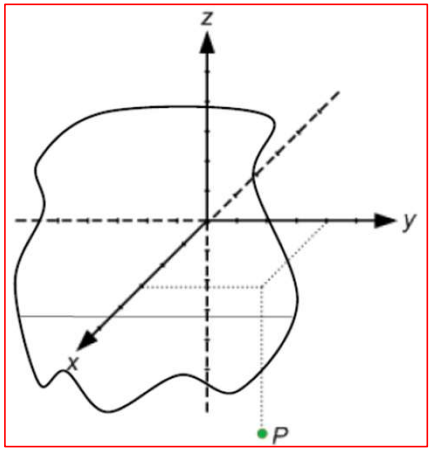
Electric Flux Density-Problem

- Example
  Calculate D at point $P(6,8,-10)$ produced by a uniform surface charge density with $\rho_s = 57.2\ \mu C/m^2$ on the plane $x = 9$.

$$\mathbf{E} = \frac{\rho_s}{2\varepsilon_0}\mathbf{a}_N \quad \Rightarrow \mathbf{D} = \frac{\rho_s}{2}\mathbf{a}_N = \frac{57.2\times10^{-6}}{2}\mathbf{a}_N = 28.6\mathbf{a}_N\ \mu C/m^2$$

At $P(6,8,-10)$,

$$\mathbf{a}_N = -\mathbf{a}_x \quad \Rightarrow \mathbf{D} = \underline{\underline{-28.6\mathbf{a}_x\ \mu C/m^2}}$$



Electric Flux Density-Practice Problem

D3.2. Calculate $\mathbf{D}$ in rectangular coordinates at point $P(2, -3, 6)$ produced by: (a) a point charge $Q_A = 55$ mC at $Q(-2, 3, -6)$; (b) a uniform line charge $\rho_{LB} = 20$ mC/m on the x axis; (c) a uniform surface charge density $\rho_{SC} = 120\,\mu\text{C/m}^2$ on the plane $z = -5$ m.

Ans. $6.38\mathbf{a}_x - 9.57\mathbf{a}_y + 19.14\mathbf{a}_z\ \mu\text{C/m}^2$; $-212\mathbf{a}_y + 424\mathbf{a}_z\ \mu\text{C/m}^2$; $60\mathbf{a}_z\ \mu\text{C/m}^2$

Gauss's Law

The electric flux passing through any closed surface is equal to the total charge enclosed by that surface.

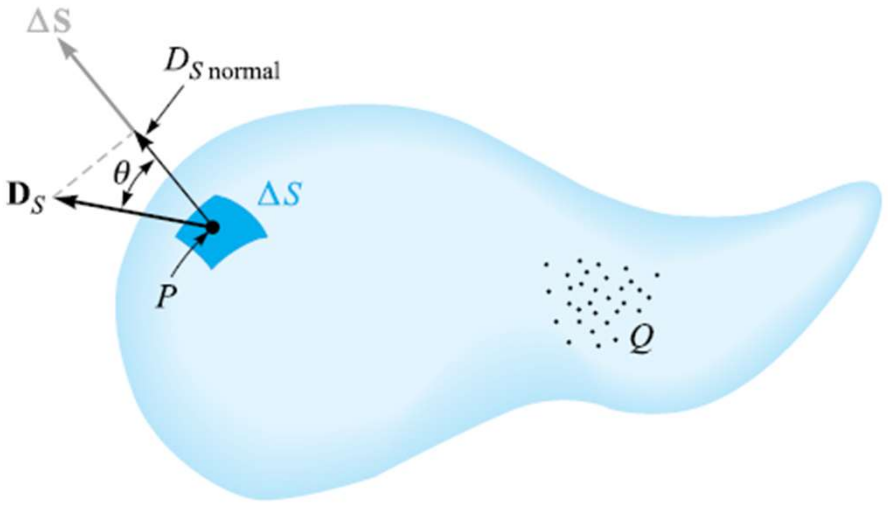


Figure 3.2 The electric flux density $\mathbf{D}_S$ at P arising from charge Q. The total flux passing through ΔS is $\mathbf{D}_S \cdot \Delta \mathbf{S}$.

The flux crossing $\Delta\mathbf{S}$ is then the product of the normal component of $\mathbf{D}_S$ and $\Delta\mathbf{S}$,

$$\Delta\Psi = \text{flux crossing } \Delta S = D_{S,\text{norm}}\Delta S = D_S \cos\theta \Delta S = \mathbf{D}_S \cdot \Delta\mathbf{S}$$

The *total* flux passing through the closed surface is obtained by adding the differential contributions crossing each surface element $\Delta\mathbf{S}$,

$$\Psi = \int d\Psi = \oint_{\substack{\text{closed}\\ \text{surface}}} \mathbf{D}_S \cdot d\mathbf{S}$$

Mathematical formulation of Gauss's law,

$$\Psi = \oint_S \mathbf{D}_S \cdot d\mathbf{S} = \text{charge enclosed} = Q$$

$$\oint_S \mathbf{D}_S \cdot d\mathbf{S} = \int_{\text{vol}} \rho_v \, dv$$

Gauss's Law

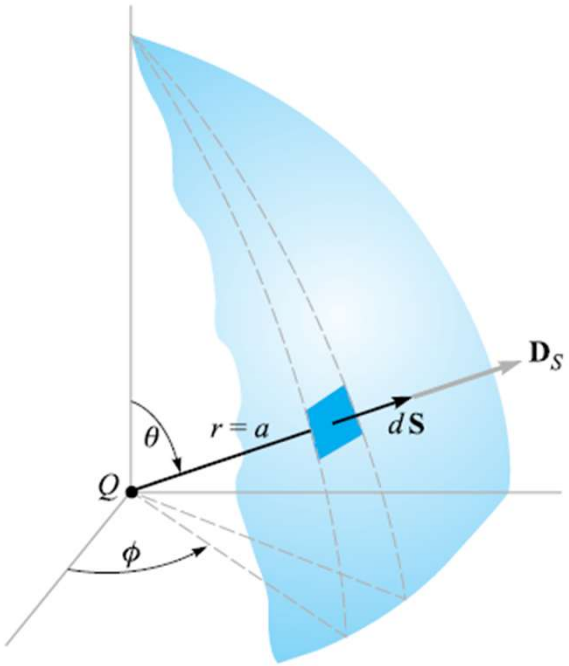


Figure 3.3 Applying Gauss's law to the field of a point charge Q on a spherical closed surface of radius a. The electric flux density $\mathbf{D}$ is everywhere normal to the spherical surface and has a constant magnitude at every point on it.

- Let a point charge Q be placed at the origin of a spherical coordinate system, and choose a closed surface as a sphere of radius a.
- The Flux Density **(D)** due to the point charge has been found to be:

$$\mathbf{D} = \frac{Q}{4\pi r^2}\mathbf{a}_r$$

At the surface of the sphere, $\mathbf{D}_S = \frac{Q}{4\pi a^2}\mathbf{a}_r$

$$dS = r^2 \sin\theta \, d\theta \, d\phi = a^2 \sin\theta \, d\theta \, d\phi$$

$$d\mathbf{S} = a^2 \sin\theta \, d\theta \, d\phi \, \mathbf{a}_r$$

The integrand is $\mathbf{D}_S \cdot d\mathbf{S} = \frac{Q}{4\pi a^2} a^2 \sin\theta \, d\theta \, d\phi \mathbf{a}_r \cdot \mathbf{a}_r = \frac{Q}{4\pi} \sin\theta \, d\theta \, d\phi$

The closed surface integral is

$$\int_{\phi=0}^{\phi=2\pi} \int_{\theta=0}^{\theta=\pi} \frac{Q}{4\pi} \sin\theta \, d\theta \, d\phi$$

$$\int_0^{2\pi} \frac{Q}{4\pi} (-\cos\theta)_0^{\pi} d\phi = \int_0^{2\pi} \frac{Q}{2\pi} d\phi = Q$$

Application of Gauss's Law

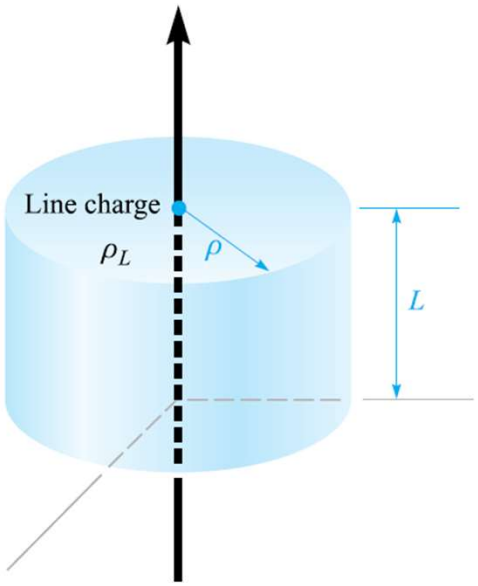


Figure 3.4 The gaussian surface for an infinite uniform line charge is a right circular cylinder of length L and radius ρ. D is constant in magnitude and everywhere perpendicular to the cylindrical surface; D is parallel to the end faces.

We apply Gauss's law,

$$Q = \oint_{\text{cyl}} \mathbf{D}_S \cdot d\mathbf{S} = D_S \int_{\text{sides}} dS + 0 \int_{\text{top}} dS + 0 \int_{\text{bottom}} dS$$

$$= D_S \int_{z=0}^{L} \int_{\phi=0}^{2\pi} \rho \, d\phi \, dz = D_S 2\pi\rho L$$

$$D_S = D_\rho = \frac{Q}{2\pi\rho L}$$

$$D_\rho = \frac{\rho_L}{2\pi\rho}$$

$$E_\rho = \frac{\rho_L}{2\pi\epsilon_0\rho}$$

Application of Gauss's Law: Co-axial Cylinder

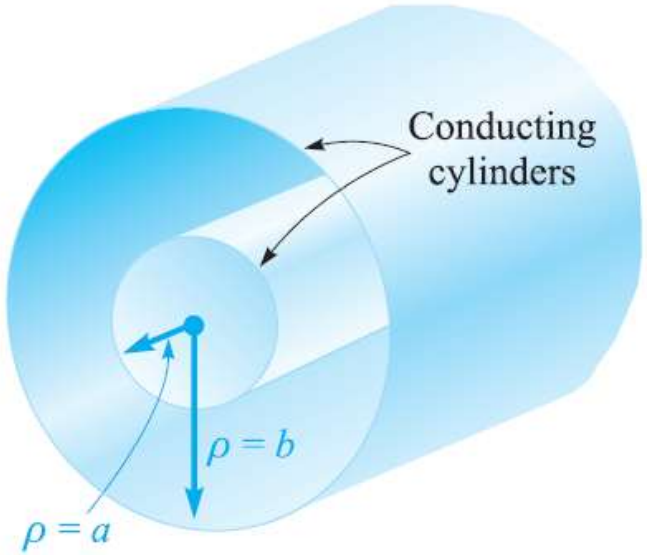


Figure 3.5 The two coaxial cylindrical conductors forming a coaxial cable provide an electric flux density within the cylinders, given by $D_\rho = a\rho_S/\rho$.

- We have two coaxial cylindrical conductors, the inner of radius ***a*** and the outer of radius ***b***, each infinite in extent (Figure 3.5).We will assume a charge distribution of ρ_S on the outer surface of the inner conductor. At Gaussian surface with cylinder of radius ***ρ***.

$$Q = D_S 2\pi\rho L$$

- The total charge on a length *L* of the inner conductor

$$Q = \int_{z=0}^{L}\int_{\phi=0}^{2\pi} \rho_S a\, d\phi\, dz = 2\pi a L \rho_S$$

$$D_S = \frac{a\rho_S}{\rho} \qquad \mathbf{D} = \frac{a\rho_S}{\rho}\mathbf{a}_\rho \qquad (a < \rho < b)$$

$$\rho_L = 2\pi a \rho_S$$

Surface charge to line charge conversion

$$\mathbf{D} = \frac{\rho_L}{2\pi\rho}\mathbf{a}_\rho$$

Application of Gauss's Law: Co-axial Cylinder-Field/Flux Outside $\rho>b$, $\rho<a$

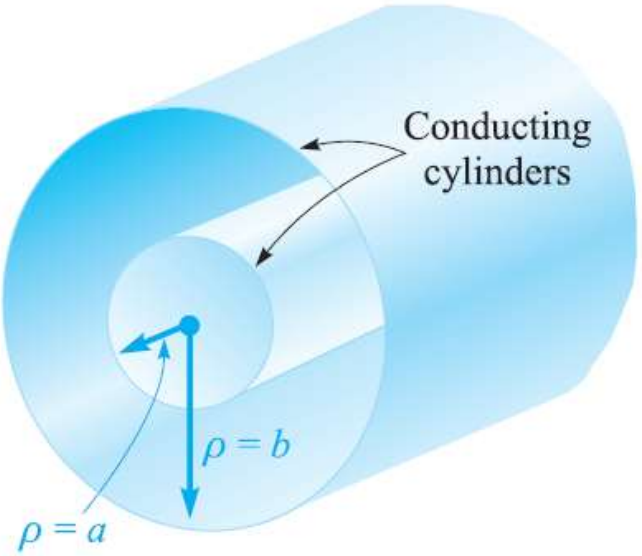


Figure 3.5 The two coaxial cylindrical conductors forming a coaxial cable provide an electric flux density within the cylinders, given by $D_\rho = a\rho_S/\rho$.

$$Q_{\text{outer cyl}} = -2\pi a L \rho_{S,\text{inner cyl}}$$

$$2\pi b L \rho_{S,\text{outer cyl}} = -2\pi a L \rho_{S,\text{inner cyl}}$$

$$\rho_{S,\text{outer cyl}} = -\frac{a}{b}\rho_{S,\text{inner cyl}}$$

$$0 = D_S 2\pi\rho L \qquad (\rho > b)$$

$$D_S = 0 \qquad (\rho > b)$$

Application of Gauss's Law: Co-axial Cylinder- Solved Problem

Let us select a 50-cm length of coaxial cable having an inner radius of 1 mm and an outer radius of 4 mm. The space between conductors is assumed to be filled with air. The total charge on the inner conductor is 30 nC. We wish to know the charge density on each conductor, and the **E** and **D** fields.

Solution. We begin by finding the surface charge density on the inner cylinder,

$$\rho_{S,\text{inner cyl}} = \frac{Q_{\text{inner cyl}}}{2\pi aL} = \frac{30 \times 10^{-9}}{2\pi(10^{-3})(0.5)} = 9.55\ \mu\text{C/m}^2$$

The negative charge density on the inner surface of the outer cylinder is

$$\rho_{S,\text{outer cyl}} = \frac{Q_{\text{outer cyl}}}{2\pi bL} = \frac{-30 \times 10^{-9}}{2\pi(4 \times 10^{-3})(0.5)} = -2.39\ \mu\text{C/m}^2$$

The internal fields may therefore be calculated easily:

$$D_\rho = \frac{a\rho_S}{\rho} = \frac{10^{-3}(9.55 \times 10^{-6})}{\rho} = \frac{9.55}{\rho}\ \text{nC/m}^2$$

and

$$E_\rho = \frac{D_\rho}{\epsilon_0} = \frac{9.55 \times 10^{-9}}{8.854 \times 10^{-12}\rho} = \frac{1079}{\rho}\ \text{V/m}$$

Both of these expressions apply to the region where $1 < \rho < 4$ mm. For $\rho < 1$ mm or $\rho > 4$ mm, **E** and **D** are zero.

Application of Gauss's Law: Differential Volume Element

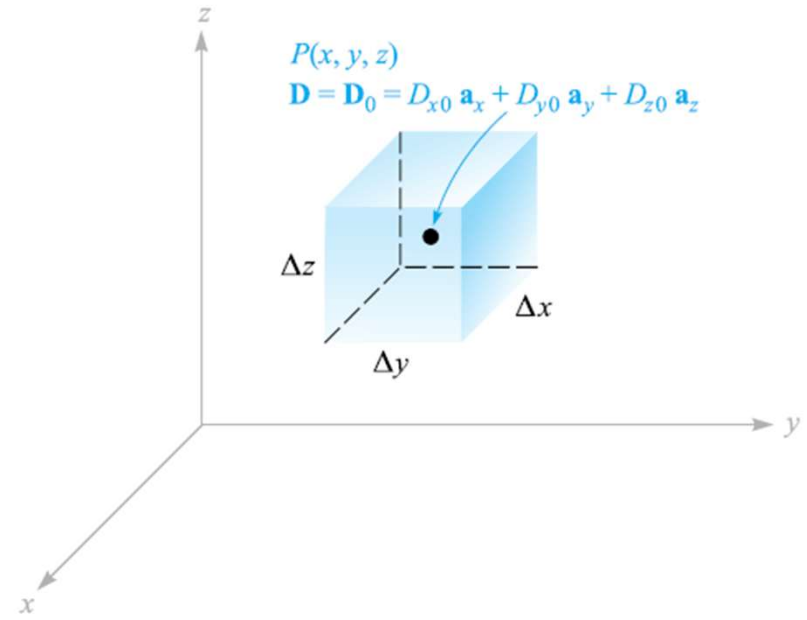


Figure 3.6 A differential-sized gaussian surface about the point *P* is used to investigate the space rate of change of **D** in the neighborhood of *P*.

$$\oint_S \mathbf{D}\cdot d\mathbf{S} \doteq \left(\frac{\partial D_x}{\partial x}+\frac{\partial D_y}{\partial y}+\frac{\partial D_z}{\partial z}\right)\Delta x\,\Delta y\,\Delta z$$

$$\oint_S \mathbf{D}\cdot d\mathbf{S} = Q \doteq \left(\frac{\partial D_x}{\partial x}+\frac{\partial D_y}{\partial y}+\frac{\partial D_z}{\partial z}\right)\Delta v$$

$$\text{Charge enclosed in volume } \Delta v \doteq \left(\frac{\partial D_x}{\partial x}+\frac{\partial D_y}{\partial y}+\frac{\partial D_z}{\partial z}\right)\times \text{volume } \Delta v$$

Find an approximate value for the total charge enclosed in an incremental volume of 10^{-9} m^3 located at the origin, if $\mathbf{D} = e^{-x}\sin y\,\mathbf{a}_x - e^{-x}\cos y\,\mathbf{a}_y + 2z\mathbf{a}_z$ C/m^2.

At the origin, the first two expressions are zero, and the last is 2. Thus, we find that the charge enclosed in a small volume element there must be approximately $2\Delta v$. If Δv is 10^{-9} m^3, then we have enclosed about 2 nC.

Application of Gauss's Law: Differential Volume Element

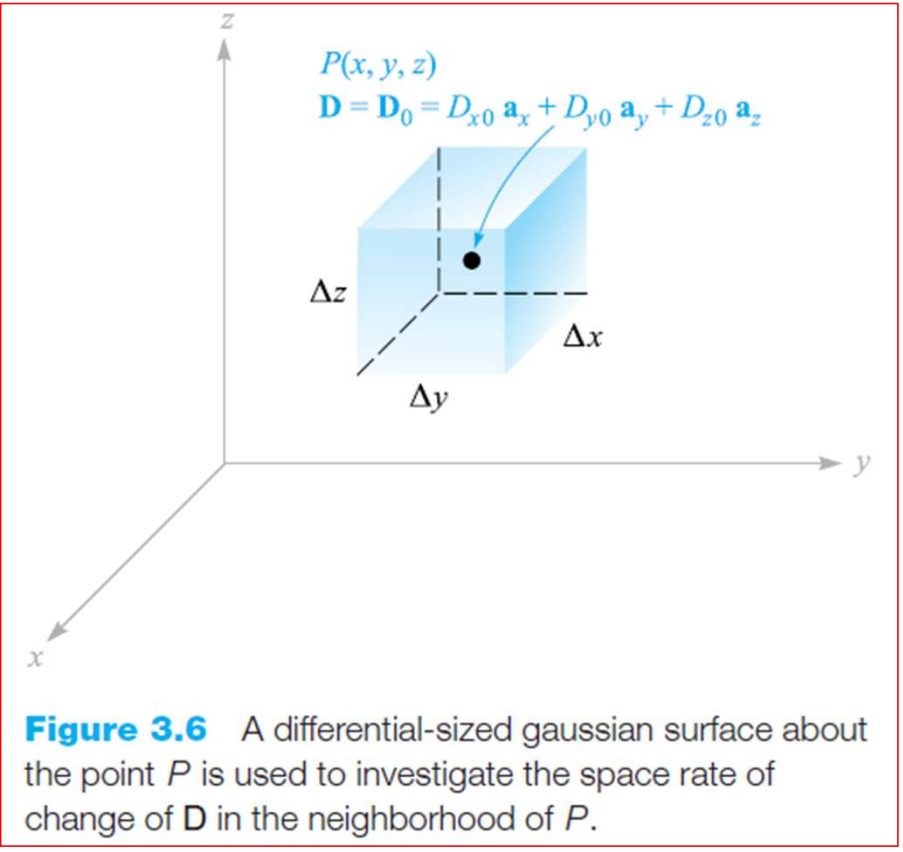
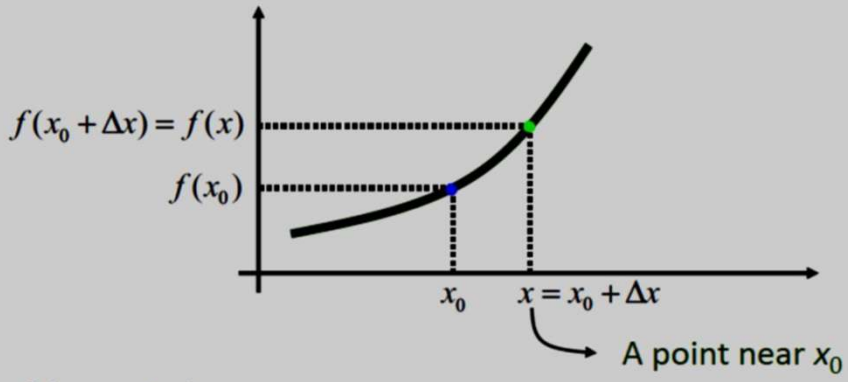


Figure 3.6 A differential-sized gaussian surface about the point *P* is used to investigate the space rate of change of **D** in the neighborhood of *P*.

Taylor's Series Expansion



$$f(x) = f(x_0 + \Delta x)$$

$$f(x) = f(x_0) + \frac{f'(x_0)}{1!}\Delta x + \frac{f''(x_0)}{2!}(\Delta x)^2 + \cdots + \frac{f^n(x_0)}{n!}(\Delta x)^n$$

Only the linear terms are used for the linearization

Application of Gauss's Law: Differential Volume Element- Problems

Example

Let $\mathbf{D} = y^2z^3\,\mathbf{a}_x + 2xyz^3\,\mathbf{a}_y + 3xy^2z^2\,\mathbf{a}_z$ nC/m^2 in free space. (*a*) Find the total electric flux passing through the surface $x = 3$, $0 \le y \le 2$, $0 \le z \le 1$ in a direction away from the origin. (*b*) Find $|\mathbf{E}|$ at $P(3,2,1)$. (*c*) Find an approximate value for the total charge contained in an incremental sphere having a radius of 2 mm centered at $P(3,2,1)$.

D3.6. In free space, let $\mathbf{D} = 8xyz^4\mathbf{a}_x + 4x^2z^4\mathbf{a}_y + 16x^2yz^3\mathbf{a}_z$ pC/m^2. (*a*) Find the total electric flux passing through the rectangular surface $z = 2$, $0 < x < 2$, $1 < y < 3$, in the $\mathbf{a}_z$ direction. (*b*) Find $\mathbf{E}$ at $P(2, -1, 3)$. (*c*) Find an approximate value for the total charge contained in an incremental sphere located at $P(2, -1, 3)$ and having a volume of 10^{-12} m^3.

Ans. 1365 pC; $-146.4\mathbf{a}_x + 146.4\mathbf{a}_y - 195.2\mathbf{a}_z$ V/m; -2.38×10^{-21} C

Solution of Example Problem

(a) $$\psi = \int_S \mathbf{D}_S \cdot d\mathbf{S}$$

$$= \int_{z=0}^{1}\int_{y=0}^{2} \left(y^2z^3\mathbf{a}_x + 2xyz^3\mathbf{a}_y + 3xy^2z^2\mathbf{a}_z\right)\cdot\left(dydz\ \mathbf{a}_x\right)\Big|_{x=3}$$

$$= \int_0^1\int_0^2 y^2z^3\,dydz$$

$$= \frac{1}{3}y^3\Big|_0^2 \frac{1}{4}z^4\Big|_0^1$$

$$= \underline{\underline{\frac{2}{3}\text{nC}}}$$

(b) $$\mathbf{D} = y^2z^3\mathbf{a}_x + 2xyz^3\mathbf{a}_y + 3xy^2z^2\mathbf{a}_z$$

$$\mathbf{D}_P = (2)^2(1)^3\mathbf{a}_x + 2(3)(2)(1)^3\mathbf{a}_y + 3(3)(2)^2(1)^2\mathbf{a}_z$$

$$= 4\mathbf{a}_x + 12\mathbf{a}_y + 36\mathbf{a}_z\ \text{nC/m}^2$$

$$|\mathbf{D}_P| = D_P = \sqrt{(4)^2 + (12)^2 + (36)^2}$$

$$= 38.158\,\text{nC/m}^2$$

$$|\mathbf{E}_P| = \frac{|\mathbf{D}_P|}{\varepsilon_0}$$

$$= \frac{38.158\ \text{nC/m}^2}{8.854\times10^{-12}}$$

$$= \underline{\underline{4.31\ \text{kV/m}}}$$

(c) $$Q = \left(\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}\right)\Delta v$$

$$Q_P = \left(\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}\right)\Big|_P \Delta v$$

$$= \left(0 + 2xz^3 + 6xy^2z\right)\Big|_{\substack{x=3\\y=2\\z=1}} \text{nC/m}^3 \times \frac{4}{3}\pi(2\times10^{-3})^3\,\text{m}^3$$

$$= \left(0 + 2(3)(1)^3 + 6(3)(2)^2(1)\right)\times\frac{4}{3}\pi(2\times10^{-3})^3\,\text{nC}$$

$$= \underline{\underline{2.61\times10^{-15}\ \text{C}}}$$

What is Divergence of a Vector Field? Divergence Theorem

$$\oint_S \mathbf{D} \cdot d\mathbf{S} \doteq \left(\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} \right) \Delta x \, \Delta y \, \Delta z$$

The way we compute the total flux for differential volume element

$$\left(\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} \right) = \lim_{\Delta v \to 0} \frac{\oint_S \mathbf{D} \cdot d\mathbf{S}}{\Delta v} = \lim_{\Delta v \to 0} \frac{Q}{\Delta v} = \rho_v$$

The Definition of Divergence, when the volume shrink to zero

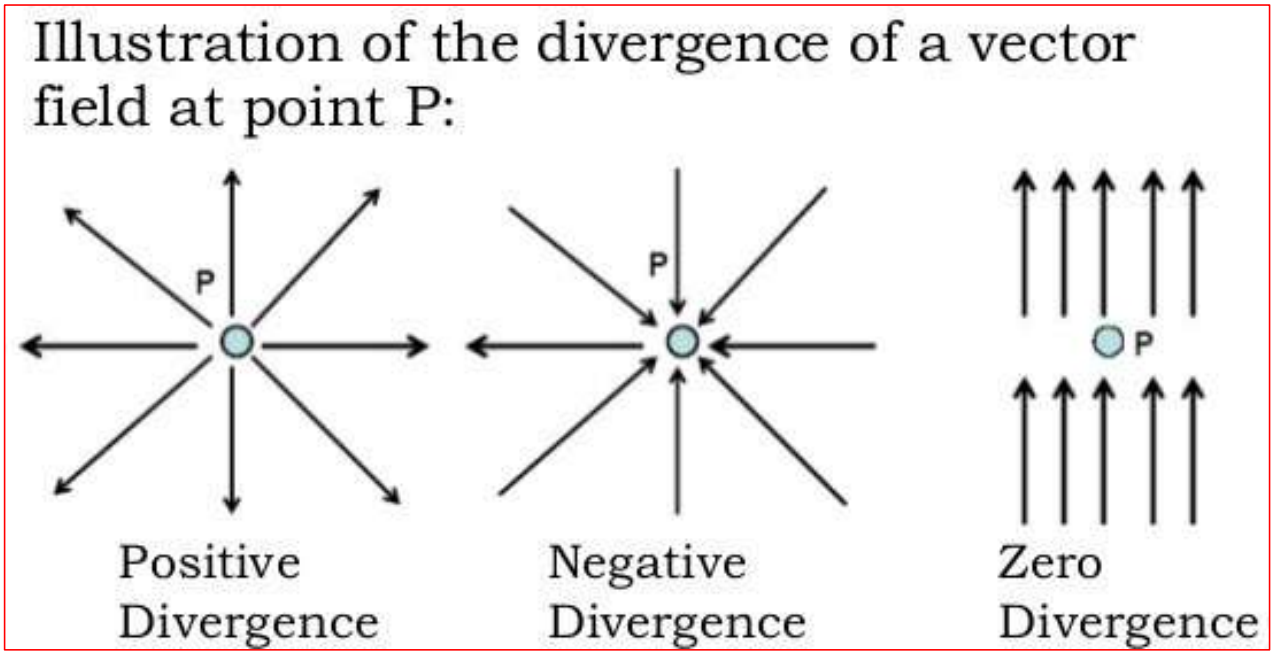
$$\left(\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \right) = \lim_{\Delta v \to 0} \frac{\oint_S \mathbf{A} \cdot d\mathbf{S}}{\Delta v}$$

Divergence for any vector field A which have flux

$$\text{Divergence of } \mathbf{A} = \text{div } \mathbf{A} = \lim_{\Delta v \to 0} \frac{\oint_S \mathbf{A} \cdot d\mathbf{S}}{\Delta v}$$

The divergence of the vector flux density $\mathbf{A}$ *is the outflow of flux from a small closed surface per unit volume as the volume shrinks to zero.*

Divergence-Visualization



- Field lines are diverging outward everywhere, like a **source** (fluid expanding from a point).
- Field lines converge inward everywhere, like a **sink** (fluid flowing into a drain).
- Field lines circulate (like a vortex), no net "source" or "sink" — fluid is incompressible.

Divergence of a Vector Field-Different Coordinate System, Del/Divergence operator

$$\text{div } \mathbf{D} = \left(\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} \right) \qquad \text{(rectangular)}$$

$$\text{div } \mathbf{D} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho D_\rho) + \frac{1}{\rho} \frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z} \qquad \text{(cylindrical)}$$

$$\text{div } \mathbf{D} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 D_r) + \frac{1}{r \sin\theta} \frac{\partial}{\partial \theta} (\sin\theta \, D_\theta) + \frac{1}{r \sin\theta} \frac{\partial D_\phi}{\partial \phi} \qquad \text{(spherical)}$$

$$\nabla = \frac{\partial}{\partial x} \mathbf{a}_x + \frac{\partial}{\partial y} \mathbf{a}_y + \frac{\partial}{\partial z} \mathbf{a}_z$$

$$\nabla \cdot \mathbf{D} = \left(\frac{\partial}{\partial x} \mathbf{a}_x + \frac{\partial}{\partial y} \mathbf{a}_y + \frac{\partial}{\partial z} \mathbf{a}_z \right) \cdot (D_x \mathbf{a}_x + D_y \mathbf{a}_y + D_z \mathbf{a}_z)$$

Divergence-Problem

Divergence

■ Example
If $D = e^{-x}\sin y\, a_x - e^{-x}\cos y\, a_y + 2z\, a_z$, find div D at the origin and $P(1,2,3)$.

$$\text{div } \mathbf{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} = -e^{-x}\sin y + e^{-x}\sin y + 2 = \underline{\underline{2}}$$

Regardless of location the divergence of D equals 2 C/m^3.

D3.7. In each of the following parts, find a numerical value for div **D** at the point specified: (*a*) $\mathbf{D} = (2xyz - y^2)\mathbf{a}_x + (x^2z - 2xy)\mathbf{a}_y + x^2y\mathbf{a}_z$ C/m^2 at $P_A(2, 3, -1)$; (*b*) $\mathbf{D} = 2\rho z^2 \sin^2\phi\, \mathbf{a}_\rho + \rho z^2 \sin 2\phi\, \mathbf{a}_\phi + 2\rho^2 z \sin^2\phi\, \mathbf{a}_z$ C/m^2 at $P_B(\rho = 2, \phi = 110°, z = -1)$; (*c*) $\mathbf{D} = 2r\sin\theta\cos\phi\, \mathbf{a}_r + r\cos\theta\cos\phi\, \mathbf{a}_\theta - r\sin\phi\, \mathbf{a}_\phi$ C/m^2 at $P_C(r = 1.5, \theta = 30°, \phi = 50°)$.

Ans. −10.00; 9.06; 1.29

Divergence-Theorem

The integral of the normal component of any vector field over a closed surface is equal to the integral of the divergence of this vector field throughout the volume enclosed by the closed surface.

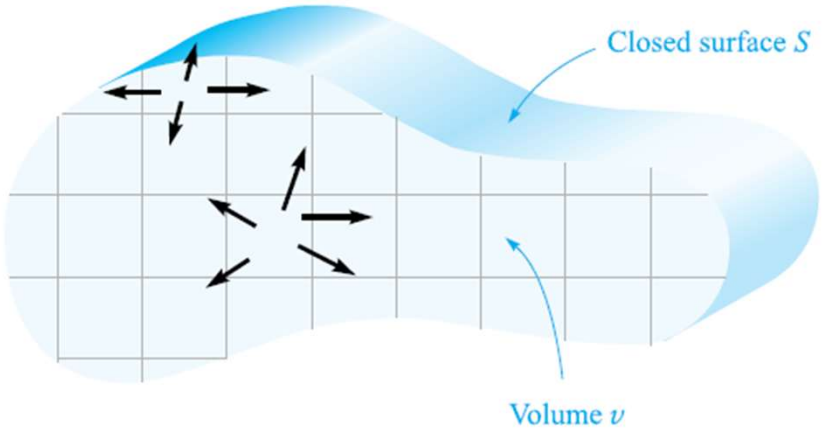


Figure 3.7 The divergence theorem states that the total flux crossing the closed surface is equal to the integral of the divergence of the flux density throughout the enclosed volume. The volume is shown here in cross section.

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = Q = \int_{\text{vol}} \rho_v dv = \int_{\text{vol}} \nabla \cdot \mathbf{D}\, dv$$

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_{\text{vol}} \nabla \cdot \mathbf{D}\, dv$$

Divergence-Theorem-Verify from the Below Problem

■ Example
Evaluate both sides of the divergence theorem for the field $\mathbf{D} = 2xy\,\mathbf{a}_x + x^2\,\mathbf{a}_y$ C/m^2 and the rectangular parallelepiped formed by the planes $x = 0$ and 1, $y = 0$ and 2, and $z = 0$ and 3.

$$\oint_S \mathbf{D}\cdot d\mathbf{S} = \int_{\text{vol}} \nabla\cdot\mathbf{D}\,dv \qquad \text{Divergence Theorem}$$

$$\oint_S \mathbf{D}_S\cdot d\mathbf{S} = \int_0^3\int_0^2 (\mathbf{D})_{x=0}\cdot(-dydz\ \mathbf{a}_x) + \int_0^3\int_0^2 (\mathbf{D})_{x=1}\cdot(dydz\ \mathbf{a}_x)$$
$$+ \int_0^3\int_0^1 (\mathbf{D})_{y=0}\cdot(-dxdz\ \mathbf{a}_y) + \int_0^3\int_0^1 (\mathbf{D})_{y=2}\cdot(dxdz\ \mathbf{a}_y)$$

But $(D_x)_{x=0} = 0, \quad (D_y)_{y=0} = (D_y)_{y=2}$

$$\oint_S \mathbf{D}_S\cdot d\mathbf{S} = \int_0^3\int_0^2 (D_x)_{x=1}\,dydz = \int_0^3\int_0^2 2y\,dydz = \underline{\underline{12\ \text{C}}}$$

$$\nabla\cdot\mathbf{D} = \frac{\partial}{\partial x}(2xy) + \frac{\partial}{\partial y}(x^2) = 2y$$

$$\int_{\text{vol}} \nabla\cdot\mathbf{D}\,dv = \int_{z=0}^3\int_{y=0}^2\int_{x=0}^1 (2y)\,dxdydz$$
$$= x\Big|_0^1\, y^2\Big|_0^2\, z\Big|_0^3$$
$$= \underline{\underline{12\ \text{C}}}$$

$$\therefore \oint_S \mathbf{D}\cdot d\mathbf{S} = \int_{\text{vol}} \nabla\cdot\mathbf{D}\,dv = 12\ \text{C}$$

Divergence-Theorem-Problem

3.22 Let $\mathbf{D} = 8\rho \sin\phi \mathbf{a}_\rho + 4\rho\cos\phi \mathbf{a}_\phi \text{C/m}^2$. (*a*) Find div **D**. (*b*) Find the volume charge density at $P(2.6, 38°, -6.1)$. (*c*) How much charge is located inside the region defined by $0 < \rho < 1.8$, $20° < \phi < 70°$, $2.4 < z < 3.1$?

Let $\mathbf{D} = 8\rho \sin\phi \, \mathbf{a}_\rho + 4\rho \cos\phi \, \mathbf{a}_\phi \text{ C/m}^2$.

a) Find div **D**: Using the divergence formula for cylindrical coordinates , we find $\nabla \cdot \mathbf{D} = \underline{12 \sin\phi}$.

$$\text{div } \mathbf{D} = \frac{1}{\rho}\frac{\partial}{\partial\rho}(\rho D_\rho) + \frac{1}{\rho}\frac{\partial D_\phi}{\partial\phi} + \frac{\partial D_z}{\partial z}$$

b) Find the volume charge density at $P(2.6, 38°, -6.1)$: Since $\rho_v = \nabla \cdot \mathbf{D}$, we evaluate the result of part *a* at this point to find $\rho_{vP} = 12 \sin 38° = \underline{7.39 \text{ C/m}^3}$.

c) How much charge is located inside the region defined by $0 < \rho < 1.8$, $20° < \phi < 70°$, $2.4 < z < 3.1$? We use

$$Q = \int_{vol} \rho_v dv = \int_{2.4}^{3.1}\int_{20°}^{70°}\int_{0}^{1.8} 12 \sin\phi\rho \, d\rho \, d\phi \, dz = -(3.1 - 2.4)12\cos\phi\Big|_{20°}^{70°} \frac{\rho^2}{2}\Big|_0^{1.8}$$

$$= \underline{8.13 \text{ C}}$$

Divergence-Theorem-Problem Verify both sides

D3.9. Given the field $\mathbf{D} = 6\rho \sin \frac{1}{2}\phi\, \mathbf{a}_\rho + 1.5\rho \cos \frac{1}{2}\phi\, \mathbf{a}_\phi$ C/m^2, evaluate both sides of the divergence theorem for the region bounded by $\rho = 2$, $\phi = 0$, $\phi = \pi$, $z = 0$, and $z = 5$.

Ans. 225; 225

Work Done for Moving a Charge Q in an Electric Field E

Suppose we wish to move a charge Q a distance $d\mathbf{L}$ in an electric field $\mathbf{E}$. The force on Q arising from the electric field is

$$\mathbf{F}_E = Q\mathbf{E} \tag{1}$$

where the subscript reminds us that this force arises from the field. The component of this force in the direction $d\mathbf{L}$ which we must overcome is

$$F_{EL} = \mathbf{F} \cdot \mathbf{a}_L = Q\mathbf{E} \cdot \mathbf{a}_L$$

where $\mathbf{a}_L$ = a unit vector in the direction of $d\mathbf{L}$.

The force that we must apply is equal and opposite to the force associated with the field,

$$F_{\text{appl}} = -Q\mathbf{E} \cdot \mathbf{a}_L$$

and the expenditure of energy is the product of the force and distance. That is, the differential work done by an external source moving charge Q is $dW = -Q\mathbf{E} \cdot \mathbf{a}_L dL$,

or

$$dW = -Q\mathbf{E} \cdot d\mathbf{L} \tag{2}$$

where we have replaced $\mathbf{a}_L dL$ by the simpler expression $d\mathbf{L}$.

Work Done for Moving a Charge Q in an Electric Field E Along a Path

Returning to the charge in the electric field, the work required to move the charge a finite distance must be determined by integrating,

$$W = -Q \int_{\text{init}}^{\text{final}} \mathbf{E} \cdot d\mathbf{L}$$

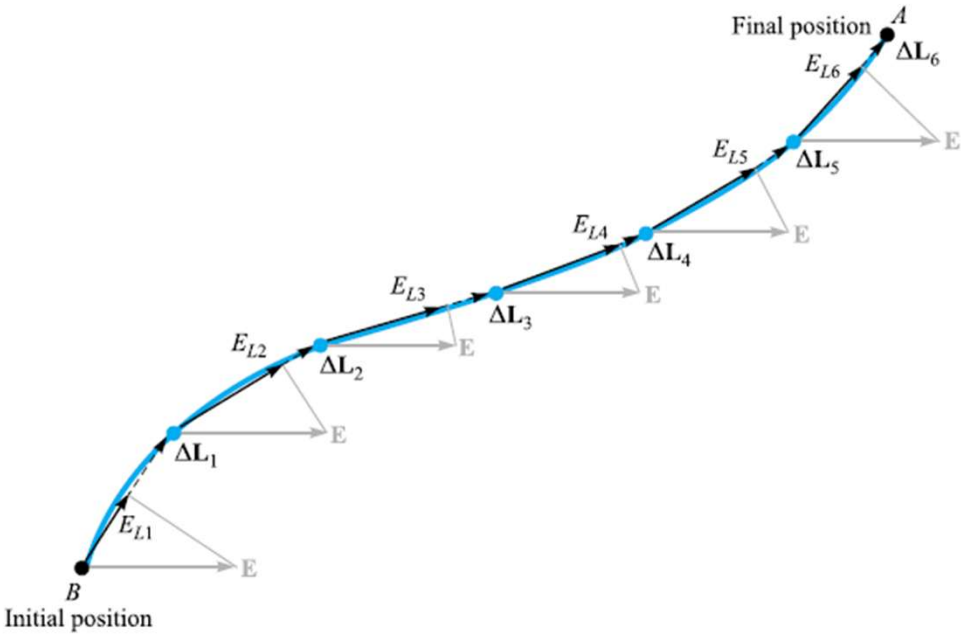


Figure 4.1 A graphical interpretation of a line integral in a uniform field. The line integral of E between points *B* and *A* is independent of the path selected, even in a nonuniform field; this result is not, in general, true for time-varying fields.

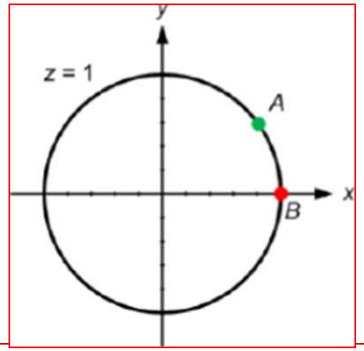
Work Done for Moving a Charge Q in an Electric Field E Along a Path-Problem-Path1-curved path

We are given the nonuniform field

$$\mathbf{E} = y\mathbf{a}_x + x\mathbf{a}_y + 2\mathbf{a}_z$$

and we are asked to determine the work expended in carrying 2C from $B(1, 0, 1)$ to $A(0.8, 0.6, 1)$ along the shorter arc of the circle

$$x^2 + y^2 = 1 \quad z = 1$$



$$\begin{aligned} W &= -Q\int_B^A \mathbf{E}\cdot d\mathbf{L} \\ &= -2\int_B^A (y\mathbf{a}_x + x\mathbf{a}_y + 2\mathbf{a}_z)\cdot(dx\,\mathbf{a}_x + dy\,\mathbf{a}_y + dz\,\mathbf{a}_z) \\ &= -2\int_1^{0.8} y\,dx - 2\int_0^{0.6} x\,dy - 4\int_1^1 dz \end{aligned}$$

$$\begin{aligned} W &= -2\int_1^{0.8}\sqrt{1-x^2}\,dx - 2\int_0^{0.6}\sqrt{1-y^2}\,dy - 0 \\ &= -\left[x\sqrt{1-x^2} + \sin^{-1}x\right]_1^{0.8} - \left[y\sqrt{1-y^2} + \sin^{-1}y\right]_0^{0.6} \\ &= -(0.48 + 0.927 - 0 - 1.571) - (0.48 + 0.644 - 0 - 0) \\ &= -0.96\,\text{J} \end{aligned}$$

$$\int\sqrt{a^2-u^2}\,du = \frac{u}{2}\sqrt{a^2-u^2} + \frac{a^2}{2}\sin^{-1}\frac{u}{a}$$

$$\begin{aligned} d\mathbf{L} &= dx\,\mathbf{a}_x + dy\,\mathbf{a}_y + dz\,\mathbf{a}_z && \text{(rectangular)} \\ d\mathbf{L} &= d\rho\,\mathbf{a}_\rho + \rho\,d\phi\mathbf{a}_\phi + dz\,\mathbf{a}_z && \text{(cylindrical)} \\ d\mathbf{L} &= dr\,\mathbf{a}_r + r\,d\theta\,\mathbf{a}_\theta + r\sin\theta\,d\phi\,\mathbf{a}_\phi && \text{(spherical)} \end{aligned}$$

Work Done for Moving a Charge Q in an Electric Field E Along a Path-Problem-Path2-Straight Line

Again find the work required to carry 2C from B to A in the same field, but this time use the straight-line path from B to A.

Solution. We start by determining the equations of the straight line. Any two of the following three equations for planes passing through the line are sufficient to define the line:

$$y - y_B = \frac{y_A - y_B}{x_A - x_B}(x - x_B)$$

$$z - z_B = \frac{z_A - z_B}{y_A - y_B}(y - y_B)$$

$$x - x_B = \frac{x_A - x_B}{z_A - z_B}(z - z_B)$$

From the first equation we have

$$y = -3(x - 1)$$

and from the second we obtain

$$z = 1$$

Thus,

$$\begin{aligned} W &= -2\int_1^{0.8} y\,dx - 2\int_0^{0.6} x\,dy - 4\int_1^1 dz \\ &= 6\int_1^{0.8} (x-1)\,dx - 2\int_0^{0.6} \left(1 - \frac{y}{3}\right) dy \\ &= -0.96\ \text{J} \end{aligned}$$

Different Problem-Path1

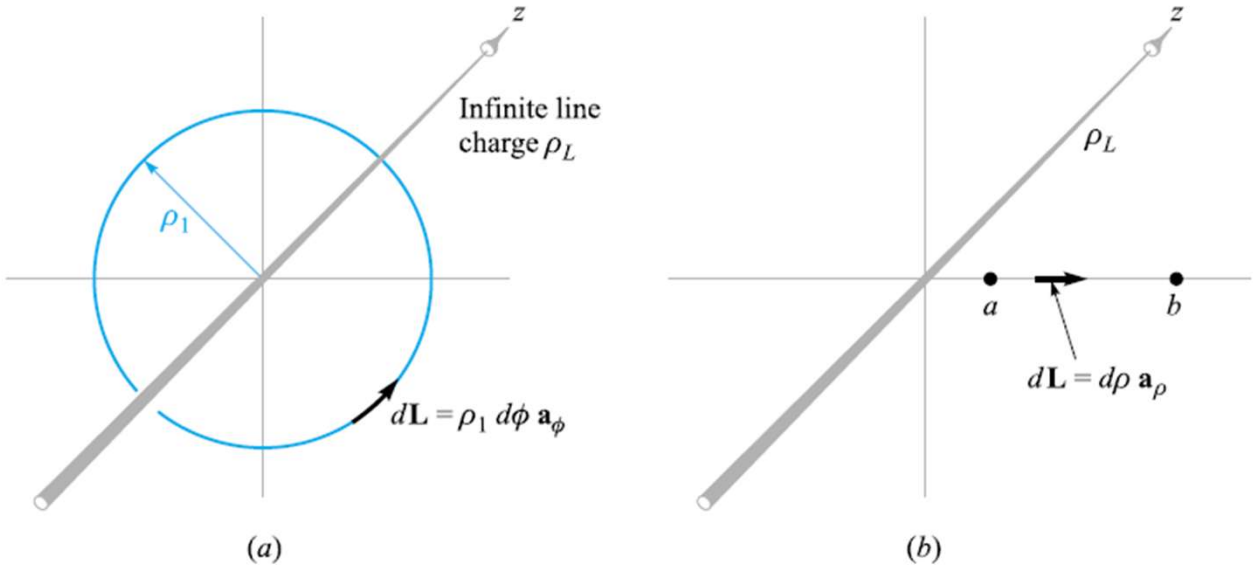


Figure 4.2 (*a*) A circular path and (*b*) a radial path along which a charge of *Q* is carried in the field of an infinite line charge. No work is expected in the former case.

The differential element $d\mathbf{L}$ is chosen in cylindrical coordinates, and the circular path selected demands that $d\rho$ and dz be zero, so $d\mathbf{L} = \rho_1\, d\phi\, \mathbf{a}_\phi$. The work is then

$$W = -Q \int_{\text{init}}^{\text{final}} \frac{\rho_L}{2\pi\epsilon_0\rho_1}\mathbf{a}_\rho \cdot \rho_1\, d\phi\, \mathbf{a}_\phi$$

$$= -Q \int_0^{2\pi} \frac{\rho_L}{2\pi\epsilon_0} d\phi\, \mathbf{a}_\rho \cdot \mathbf{a}_\phi = 0$$

Different Problem-Path2

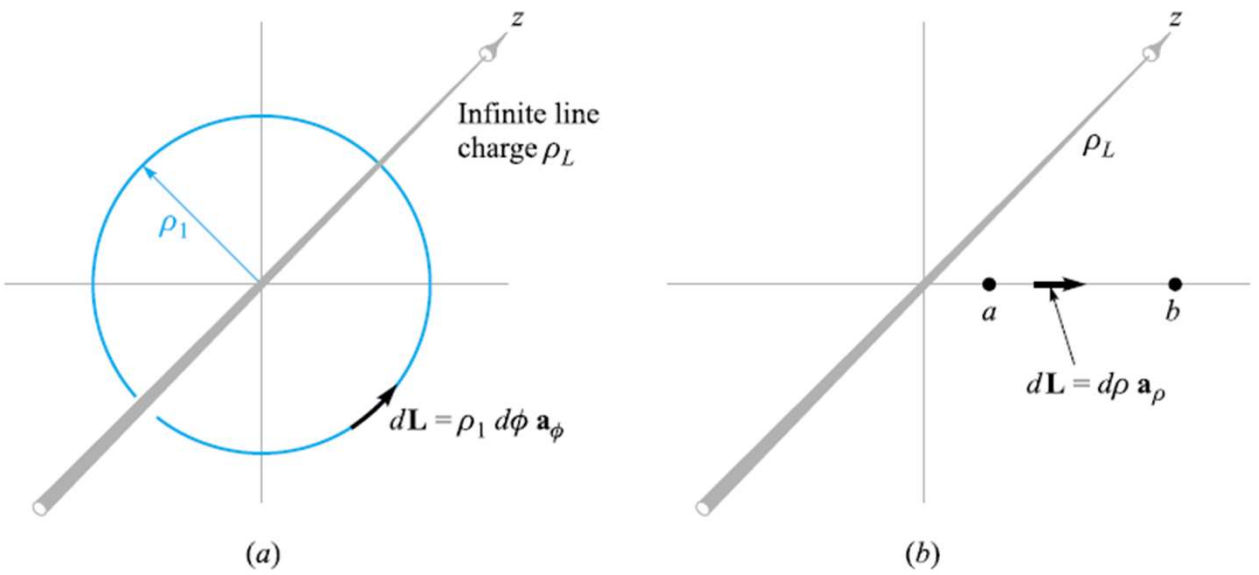


Figure 4.2 (*a*) A circular path and (*b*) a radial path along which a charge of *Q* is carried in the field of an infinite line charge. No work is expected in the former case.

We will now carry the charge from $\rho = a$ to $\rho = b$ along a radial path (Figure 4.2*b*). Here $d\mathbf{L} = d\rho\, \mathbf{a}_\rho$ and

$$W = -Q \int_{\text{init}}^{\text{final}} \frac{\rho_L}{2\pi\epsilon_0\rho} \mathbf{a}_\rho \cdot d\rho\, \mathbf{a}_\rho = -Q \int_a^b \frac{\rho_L}{2\pi\epsilon_0} \frac{d\rho}{\rho}$$

or

$$W = -\frac{Q\rho_L}{2\pi\epsilon_0} \ln \frac{b}{a}$$

Because b is larger than a, $\ln(b/a)$ is positive, and the work done is negative, indicating that the external source that is moving the charge receives energy.

Different Problem-Path2*

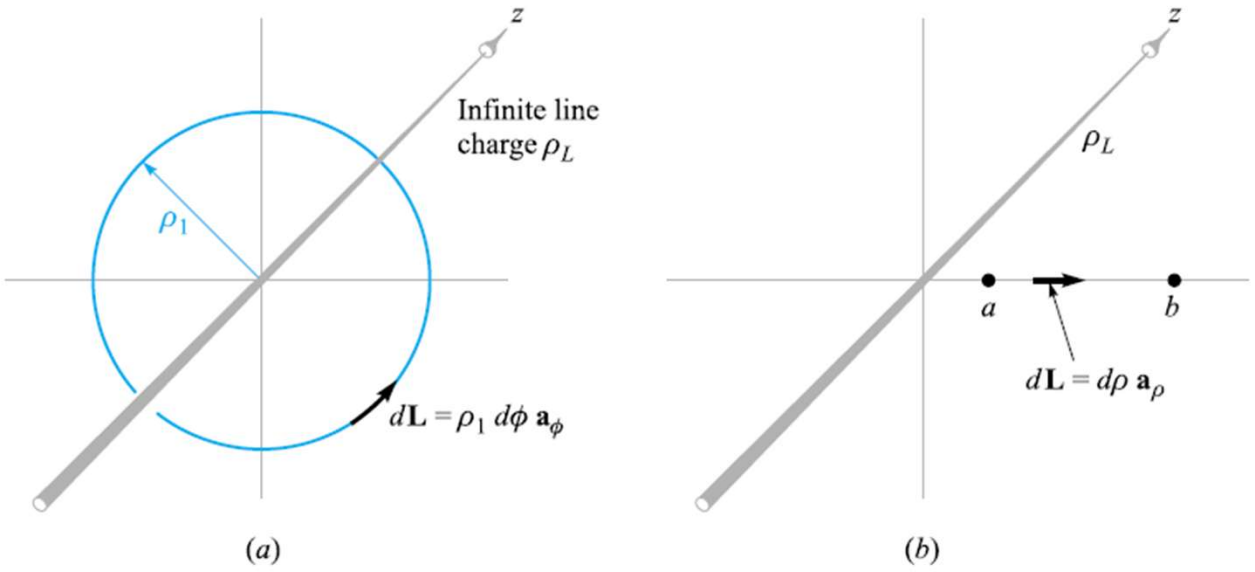


Figure 4.2 (*a*) A circular path and (*b*) a radial path along which a charge of *Q* is carried in the field of an infinite line charge. No work is expected in the former case.

One of the pitfalls in evaluating line integrals is a tendency to use too many minus signs when a charge is moved in the direction of a *decreasing* coordinate value. This is taken care of completely by the limits on the integral, and no misguided attempt should be made to change the sign of $d\mathbf{L}$. Suppose we carry Q from b to a (Figure 4.2*b*). We still have $d\mathbf{L} = d\rho\, \mathbf{a}_\rho$ and show the different direction by recognizing $\rho = b$ as the initial point and $\rho = a$ as the final point,

$$W = -Q \int_b^a \frac{\rho_L}{2\pi\epsilon_0} \frac{d\rho}{\rho} = \frac{Q\rho_L}{2\pi\epsilon_0} \ln \frac{b}{a}$$

This is the negative of the previous answer and is obviously correct.

Practice Problem

D4.2. Calculate the work done in moving a 4-C charge from $B(1, 0, 0)$ to $A(0, 2, 0)$ along the path $y = 2 - 2x$, $z = 0$ in the field $\mathbf{E} = (a)\ 5\mathbf{a}_x$ V/m; $(b)\ 5x\mathbf{a}_x$ V/m; $(c)\ 5x\mathbf{a}_x + 5y\mathbf{a}_y$ V/m.

Ans. 20 J; 10 J; −30 J

Potential Different and Electric Potential

We are now ready to define a new concept from the expression for the work done by an external source in moving a charge Q from one point to another in an electric field $\mathbf{E}$, "Potential difference and work."

$$W = -Q \int_{\text{init}}^{\text{final}} \mathbf{E} \cdot d\mathbf{L}$$

In much the same way as we defined the electric field intensity as the force on a *unit* test charge, we now define *potential difference* V as the work done (by an external source) in moving a *unit* positive charge from one point to another in an electric field,

$$\text{Potential difference} = V = -\int_{\text{init}}^{\text{final}} \mathbf{E} \cdot d\mathbf{L} \qquad (9)$$

Potential Different and Electric Potential

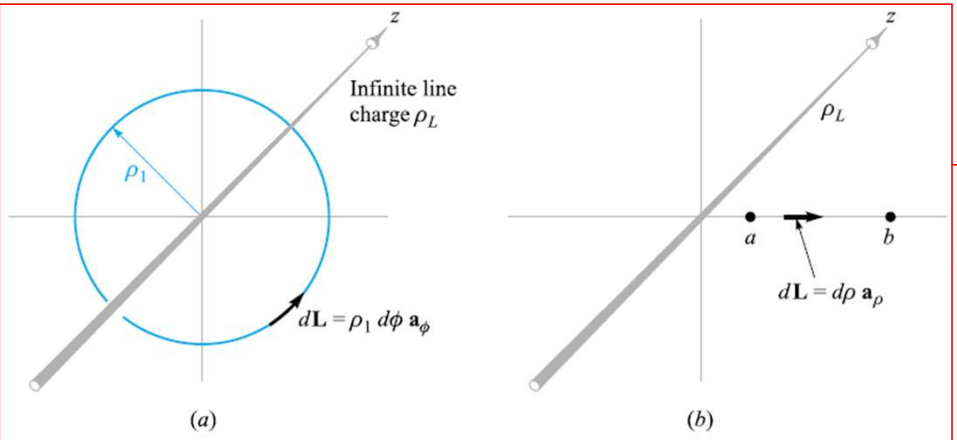


Figure 4.2 (*a*) A circular path and (*b*) a radial path along which a charge of *Q* is carried in the field of an infinite line charge. No work is expected in the former case.

Potential difference is measured in joules per coulomb, for which the *volt* is defined as a more common unit, abbreviated as V. Hence the potential difference between points *A* and *B* is

$$V_{AB} = -\int_B^A \mathbf{E} \cdot d\mathbf{L} \text{ V} \tag{10}$$

and V_{AB} is positive if work is done in carrying the positive charge from *B* to *A*.

From the line-charge example of Section 4.2 we found that the work done in taking a charge *Q* from $\rho = b$ to $\rho = a$ was

$$W = \frac{Q\rho_L}{2\pi\epsilon_0} \ln \frac{b}{a}$$

Thus, the potential difference between points at $\rho = a$ and $\rho = b$ is

$$V_{ab} = \frac{W}{Q} = \frac{\rho_L}{2\pi\epsilon_0} \ln \frac{b}{a} \tag{11}$$

Potential Difference and Electric Potential

We can try out this definition by finding the potential difference between points A and B at radial distances r_A and r_B from a point charge Q. Choosing an origin at Q,

$$\mathbf{E} = E_r \mathbf{a}_r = \frac{Q}{4\pi\epsilon_0 r^2}\mathbf{a}_r$$

and

$$d\mathbf{L} = dr\,\mathbf{a}_r$$

we have

$$V_{AB} = -\int_B^A \mathbf{E}\cdot d\mathbf{L} = -\int_{r_B}^{r_A} \frac{Q}{4\pi\epsilon_0 r^2}\,dr = \frac{Q}{4\pi\epsilon_0}\left(\frac{1}{r_A} - \frac{1}{r_B}\right) \qquad (12)$$

If $r_B > r_A$, the potential difference V_{AB} is positive, indicating that energy is expended by the external source in bringing the positive charge from r_B to r_A. This agrees with the physical picture showing the two like charges repelling each other.

If the potential at point A is V_A and that at B is V_B, then

$$V_{AB} = V_A - V_B \qquad (13)$$

where we necessarily agree that V_A and V_B shall have the same zero reference point.

Work Done around a Closed Path

This result is often stated concisely by recognizing that no work is done in carrying the unit charge around any *closed path*, or

$$\oint \mathbf{E} \cdot d\mathbf{L} = 0 \tag{20}$$

A small circle is placed on the integral sign to indicate the closed nature of the path. This symbol also appeared in the formulation of Gauss's law, where a closed *surface* integral was used.

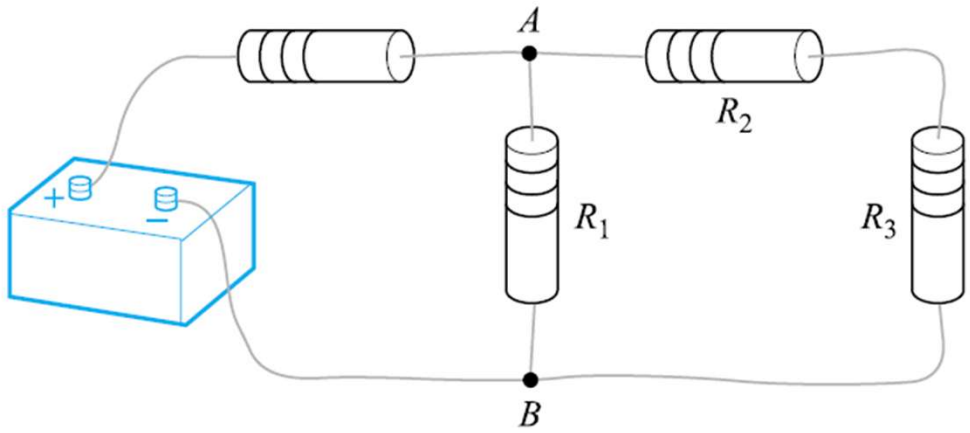


Figure 4.4 A simple dc-circuit problem that must be solved by applying $\oint \mathbf{E} \cdot d\mathbf{L} = 0$ in the form of Kirchhoff's voltage law.

$$\mathbf{E} = -\frac{dV}{dN}\mathbf{a}_N = -\text{grad } V$$

$$\mathbf{E} = -\nabla V$$

$$\mathbf{E} = -\left(\frac{\partial V}{\partial x}\mathbf{a}_x + \frac{\partial V}{\partial y}\mathbf{a}_y + \frac{\partial V}{\partial z}\mathbf{a}_z\right)$$

- **Note: Gradient of a scalar is a vector.**

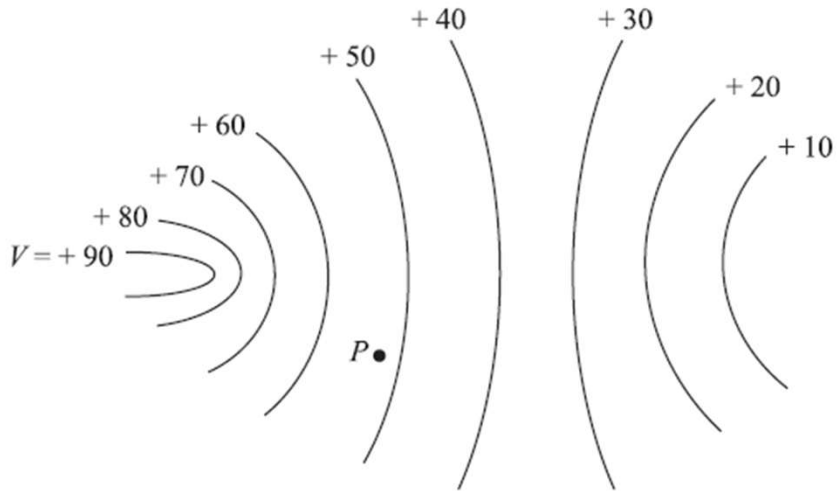


Figure 4.6 A potential field is shown by its equipotential surfaces. At any point the E field is normal to the equipotential surface passing through that point and is directed toward the more negative surfaces.

$$\nabla V = \frac{\partial V}{\partial \rho}\mathbf{a}_\rho + \frac{1}{\rho}\frac{\partial V}{\partial \phi}\mathbf{a}_\phi + \frac{\partial V}{\partial z}\mathbf{a}_z$$

Cylindrical

$$\nabla V = \frac{\partial V}{\partial r}\mathbf{a}_r + \frac{1}{r}\frac{\partial V}{\partial \theta}\mathbf{a}_\theta + \frac{1}{r\sin\theta}\frac{\partial V}{\partial \phi}\mathbf{a}_\phi$$

Spherical

■ Example

Given the potential field, $V = 2x^2y - 5z$, and a point $P(-4,3,6)$, find V, E, direction of E, D, and ρ_V.

$$V_P = 2(-4)^2(3) - 5(6) = \underline{66 \text{ V}}$$

$$\mathbf{E} = -\nabla V = -\left(\frac{\partial V}{\partial x}\mathbf{a}_x + \frac{\partial V}{\partial y}\mathbf{a}_y + \frac{\partial V}{\partial z}\mathbf{a}_z \right) = -4xy\mathbf{a}_x - 2x^2\mathbf{a}_y + 5\mathbf{a}_z$$

$$\mathbf{E}_P = -4(-4)(3)\mathbf{a}_x - 2(-4)^2\mathbf{a}_y + 5\mathbf{a}_z = \underline{48\mathbf{a}_x - 32\mathbf{a}_y + 5\mathbf{a}_z \text{ V/m}}$$

$$\mathbf{a}_{E,P} = \frac{\mathbf{E}_P}{|\mathbf{E}_P|}$$

$$\mathbf{D}_P = \varepsilon_0 \mathbf{E}_P = \underline{425\mathbf{a}_x - 283.3\mathbf{a}_y + 44.27\mathbf{a}_z \text{ pC/m}^3}$$

$$\rho_v = \text{div } \mathbf{D} = \text{div } \varepsilon_0 \mathbf{E} = (8.854 \times 10^{-12})(-4y) = -35.42y \text{ pC/m}^3$$

$$\text{At } P,\ \rho_v = -35.42(3) \text{ pC/m}^3 = \underline{-106.26 \text{ pC/m}^3}$$

D4.8. Given the potential field in cylindrical coordinates, $V = \frac{100}{z^2+1}\rho\cos\phi$ V, and point P at $\rho = 3$ m, $\phi = 60°$, $z = 2$ m, find values at P for (*a*) V; (*b*) **E**; (*c*) E; (*d*) dV/dN; (*e*) $\mathbf{a}_N$; (*f*) ρ_v in free space.

$$\nabla V = \frac{\partial V}{\partial \rho}\mathbf{a}_\rho + \frac{1}{\rho}\frac{\partial V}{\partial \phi}\mathbf{a}_\phi + \frac{\partial V}{\partial z}\mathbf{a}_z$$

Cylindrical

$$\nabla V = \frac{\partial V}{\partial r}\mathbf{a}_r + \frac{1}{r}\frac{\partial V}{\partial \theta}\mathbf{a}_\theta + \frac{1}{r\sin\theta}\frac{\partial V}{\partial \phi}\mathbf{a}_\phi$$

Spherical

ENERGY DENSITY IN THE ELECTROSTATIC FIELD

- Bringing a positive charge from infinity into the field of another positive charge requires work, the work being done by the external source moving the charge.
- Energy must be conserved, and the energy expended in bringing this charge into position now represents potential energy.
- In order to find the potential energy present in a system of charges, we must find the work done by an external source in positioning the charges.

Work to position $Q_2 = Q_2 V_{2,1}$

Work to position $Q_3 = Q_3 V_{3,1} + Q_3 V_{3,2}$

Work to position $Q_4 = Q_4 V_{4,1} + Q_4 V_{4,2} + Q_4 V_{4,3}$

$V_{2,1}$ is the potential at the location of Q_2 due to Q_1.

Total positioning work = potential energy of field

$$= W_E = Q_2 V_{2,1} + Q_3 V_{3,1} + Q_3 V_{3,2} + Q_4 V_{4,1} + Q_4 V_{4,2} + Q_4 V_{4,3} + \cdots$$

ENERGY DENSITY IN THE ELECTROSTATIC FIELD

Total positioning work = potential energy of field

$$= W_E = Q_2 V_{2,1} + Q_3 V_{3,1} + Q_3 V_{3,2} + Q_4 V_{4,1} + Q_4 V_{4,2} + Q_4 V_{4,3} + \cdots$$

$$Q_3 V_{3,1} = Q_3 \frac{Q_1}{4\pi\epsilon_0 R_{13}} = Q_1 \frac{Q_3}{4\pi\epsilon_0 R_{31}}$$

$$W_E = Q_1 V_{1,2} + Q_1 V_{1,3} + Q_2 V_{2,3} + Q_1 V_{1,4} + Q_2 V_{2,4} + Q_3 V_{3,4} + \cdots$$

ENERGY DENSITY IN THE ELECTROSTATIC FIELD

$$W_E = Q_2 V_{2,1} + Q_3 V_{3,1} + Q_3 V_{3,2} + Q_4 V_{4,1} + Q_4 V_{4,2} + Q_4 V_{4,3} + \cdots$$

$$W_E = Q_1 V_{1,2} + Q_1 V_{1,3} + Q_2 V_{2,3} + Q_1 V_{1,4} + Q_2 V_{2,4} + Q_3 V_{3,4} + \cdots$$

$$\begin{aligned} 2W_E = {} & Q_1(V_{1,2} + V_{1,3} + V_{1,4} + \cdots) \\ & + Q_2(V_{2,1} + V_{2,3} + V_{2,4} + \cdots) \\ & + Q_3(V_{3,1} + V_{3,2} + V_{3,4} + \cdots) \\ & + \cdots \end{aligned}$$

$$V_{1,2} + V_{1,3} + V_{1,4} + \cdots = V_1$$

V_1 is the potential at the location of Q_1 due to the presence of $Q_2, Q_3, \ldots$.

$$W_E = \frac{1}{2}(Q_1 V_1 + Q_2 V_2 + Q_3 V_3 + \cdots) = \frac{1}{2}\sum_{m=1}^{m=N} Q_m V_m$$

$$W_E = \frac{1}{2}\int_{\text{vol}} \rho_v V \, dv$$

ENERGY DENSITY IN THE ELECTROSTATIC FIELD

$$W_E = \frac{1}{2}\int_{\text{vol}} \rho_v V\,dv = \frac{1}{2}\int_{\text{vol}} (\nabla\cdot\mathbf{D})V\,dv$$

$$= \frac{1}{2}\int_{\text{vol}} [\nabla\cdot(V\mathbf{D}) - \mathbf{D}\cdot(\nabla V)]\,dv$$

(Vector Identity)

$$\nabla\cdot(V\mathbf{D}) \equiv V(\nabla\cdot\mathbf{D}) + \mathbf{D}\cdot(\nabla V)$$

$$W_E = \frac{1}{2}\oint_S (V\mathbf{D})\cdot d\mathbf{S} - \frac{1}{2}\int_{\text{vol}} \mathbf{D}\cdot(\nabla V)\,dv$$

(Divergence theorem)

(First term is zero in expression)

$$W_E = \frac{1}{2}\int_{\text{vol}} \mathbf{D}\cdot\mathbf{E}\,dv = \frac{1}{2}\int_{\text{vol}} \epsilon_0 E^2\,dv$$

ENERGY DENSITY IN THE ELECTROSTATIC FIELD

$$W_E = \frac{1}{2}\int_{\text{vol}} \mathbf{D}\cdot\mathbf{E}\, dv$$

and write it on a differential basis,

$$dW_E = \tfrac{1}{2}\mathbf{D}\cdot\mathbf{E}\, dv$$

or

$$\frac{dW_E}{dv} = \tfrac{1}{2}\mathbf{D}\cdot\mathbf{E}$$

D4.11. Find the energy stored in free space for the region 2 mm $< r <$ 3 mm, $0 < \theta < 90°$, $0 < \phi < 90°$, given the potential field $V =:$ (*a*) $\dfrac{200}{r}$ V; (*b*) $\dfrac{300\cos\theta}{r^2}$ V.

Boundary Condition at Conductor-Free Space

- If the field exists in a region consisting of two different media, the conditions that the field must satisfy at the interface separating the media are called boundary conditions.

- Conductor and Free space

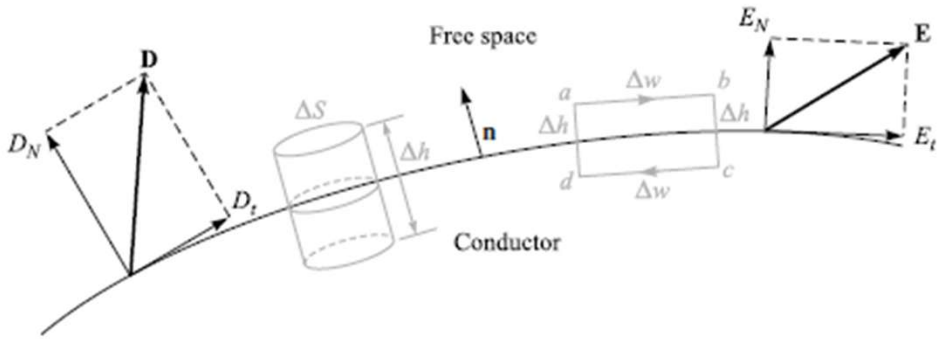


Figure 5.4 An appropriate closed path and gaussian surface are used to determine boundary conditions at a boundary between a conductor and free space; $E_t = 0$ and $D_N = \rho_S$.

Boundary Condition at Conductor-Free Space

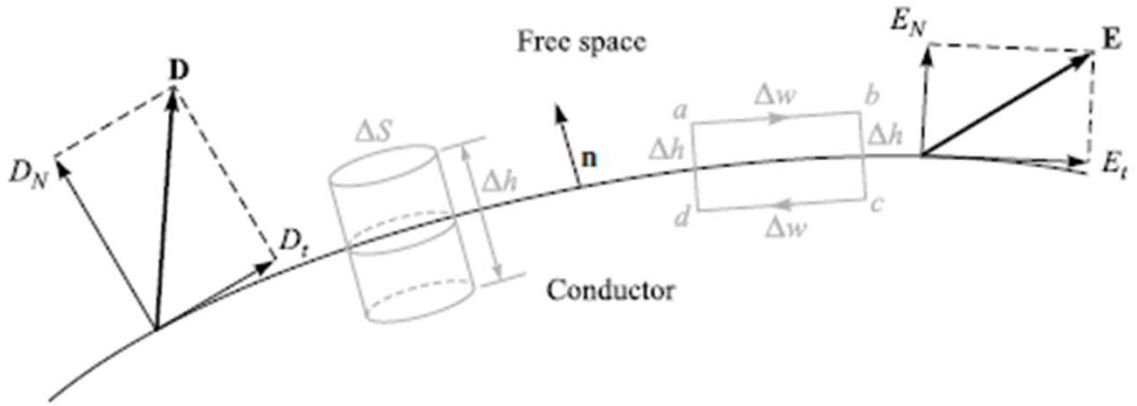


Figure 5.4 An appropriate closed path and gaussian surface are used to determine boundary conditions at a boundary between a conductor and free space; $E_t = 0$ and $D_N = \rho_S$.

$$\oint \mathbf{E} \cdot d\mathbf{L} = 0$$

$$\int_a^b + \int_b^c + \int_c^d + \int_d^a = 0$$

$$E_t \Delta w - E_{N,\text{at } b} \tfrac{1}{2}\Delta h + E_{N,\text{at } a} \tfrac{1}{2}\Delta h = 0$$

$$E_t \Delta w = 0 \text{ and, therefore, } E_t = 0.$$

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = Q \qquad \int_{\text{top}} + \int_{\text{bottom}} + \int_{\text{sides}} = Q$$

$$D_N \Delta S = Q = \rho_S \Delta S$$

$$D_t = E_t = 0$$

$$D_N = \epsilon_0 E_N = \rho_S$$

Boundary Condition at Conductor-Free Space

1. The static electric field intensity inside a conductor is zero.
2. The static electric field intensity at the surface of a conductor is everywhere directed normal to that surface.
3. The conductor surface is an equipotential surface.

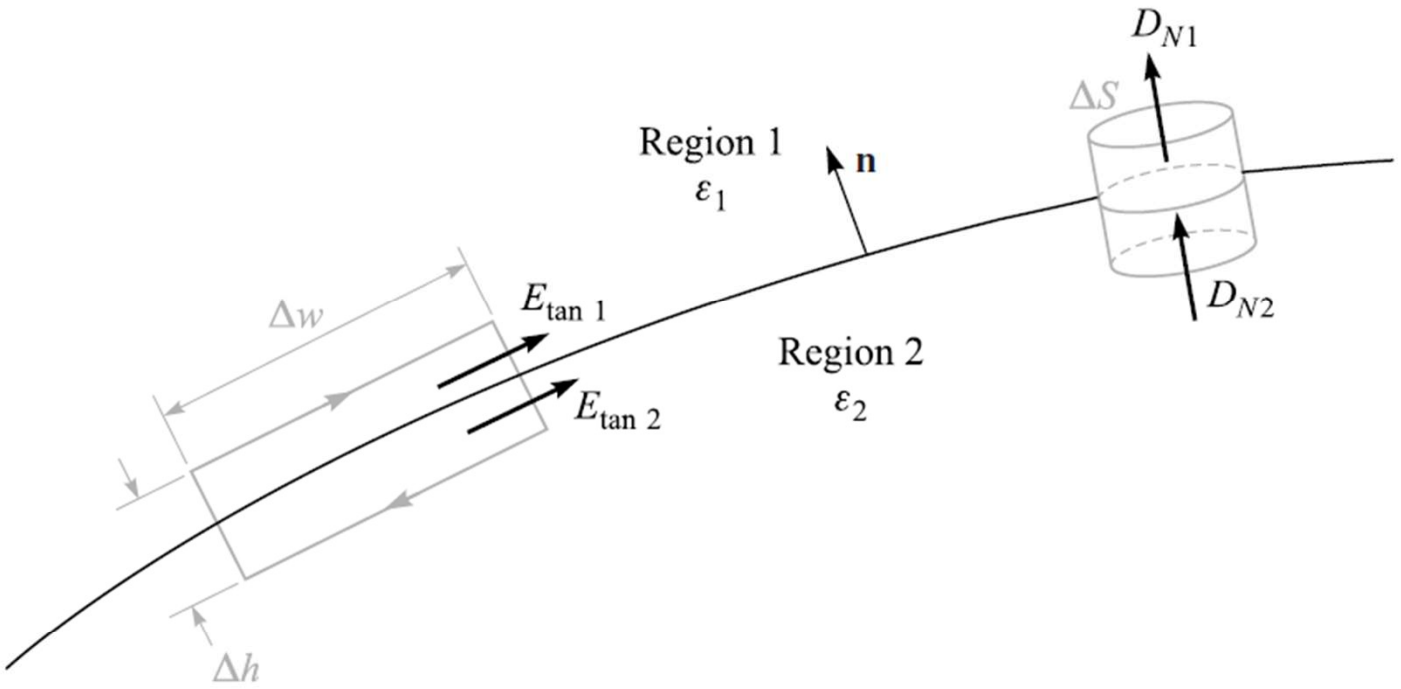


Figure 5.10 The boundary between perfect dielectrics of permittivities ϵ_1 and ϵ_2. The continuity of D_N is shown by the gaussian surface on the right, and the continuity of $E_{\tan}$ is shown by the line integral about the closed path at the left.

BOUNDARY CONDITIONS FOR PERFECT DIELECTRIC MATERIALS

$$\oint \mathbf{E} \cdot d\mathbf{L} = 0$$

$$E_{\tan 1}\,\Delta w - E_{\tan 2}\,\Delta w = 0$$

$$\boxed{E_{\tan 1} = E_{\tan 2}}$$

$$\frac{D_{\tan 1}}{D_{\tan 2}} = \frac{\epsilon_1}{\epsilon_2}$$

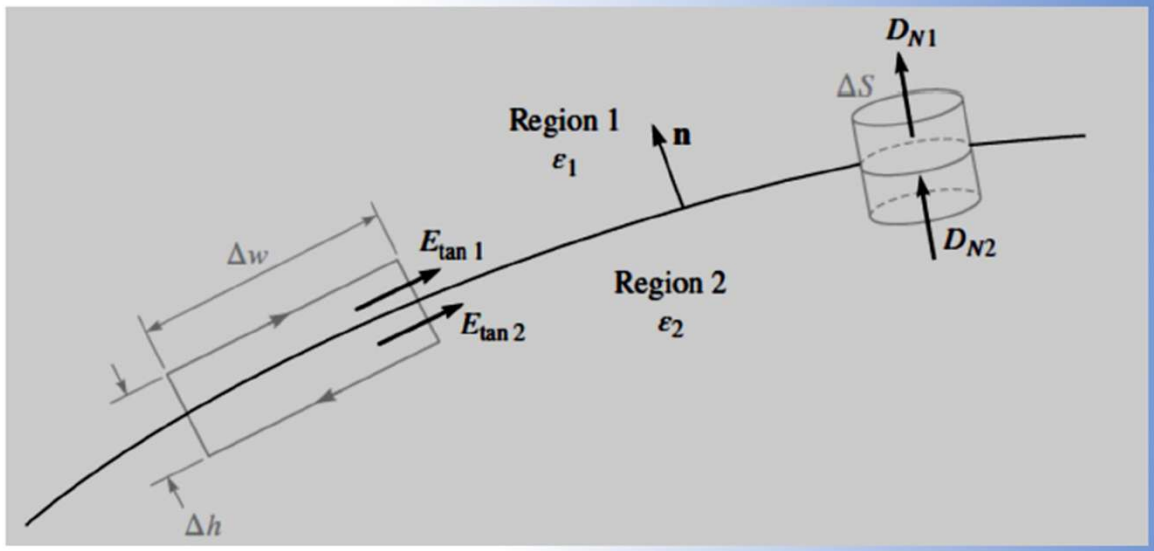
$$D_{N1}\,\Delta S - D_{N2}\,\Delta S = \Delta Q = \rho_S \Delta S$$

$$\boxed{D_{N1} - D_{N2} = \rho_S}$$

If $\rho_s = 0$

$$\boxed{D_{N1} = D_{N2}}$$

$$\epsilon_1 E_{N1} = \epsilon_2 E_{N2}$$



$$D_{N1} = D_{N2}$$

$$\frac{D_{\tan 1}}{D_{\tan 2}} = \frac{\epsilon_1}{\epsilon_2}$$

$$D_{N1} = D_1 \cos\theta_1 = D_2 \cos\theta_2 = D_{N2}$$

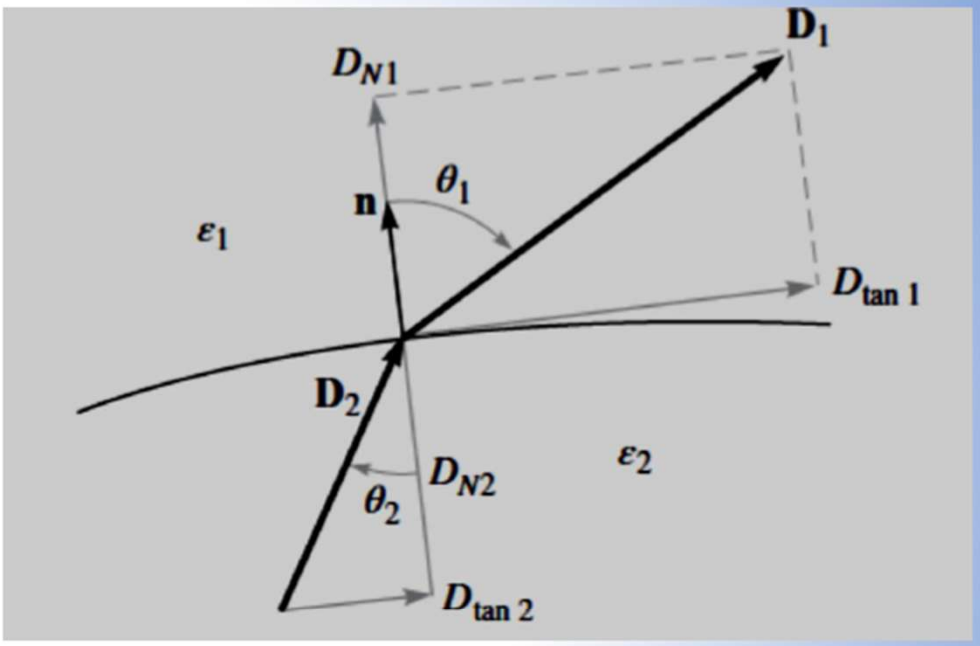
$$\frac{D_{\tan 1}}{D_{\tan 2}} = \frac{D_1 \sin\theta_1}{D_2 \sin\theta_2} = \frac{\epsilon_1}{\epsilon_2}$$

$$\epsilon_2 D_1 \sin\theta_1 = \epsilon_1 D_2 \sin\theta_2$$

$$\frac{\tan\theta_1}{\tan\theta_2} = \frac{\epsilon_1}{\epsilon_2}$$

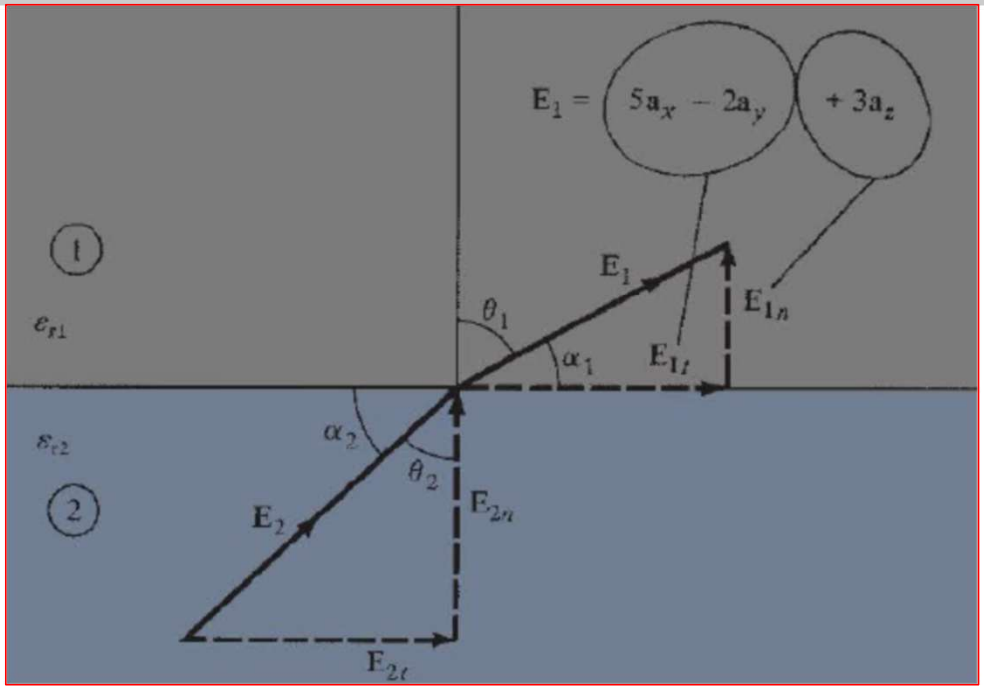
$$D_2 = D_1 \sqrt{\cos^2\theta_1 + \left(\frac{\epsilon_2}{\epsilon_1}\right)^2 \sin^2\theta_1}$$

$$E_2 = E_1 \sqrt{\sin^2\theta_1 + \left(\frac{\epsilon_1}{\epsilon_2}\right)^2 \cos^2\theta_1}$$



Two extensive homogeneous isotropic dielectrics meet on plane $z = 0$. For $z > 0$, $\varepsilon_{r1} = 4$ and for $z < 0$, $\varepsilon_{r2} = 3$. A uniform electric field $\mathbf{E}_1 = 5\mathbf{a}_x - 2\mathbf{a}_y + 3\mathbf{a}_z$ kV/m exists for $z \geq 0$. Find

(a) $\mathbf{E}_2$ for $z \leq 0$
(b) The angles $\mathbf{E}_1$ and $\mathbf{E}_2$ make with the interface
(c) The energy densities (in J/m^3) in both dielectrics
(d) The energy within a cube of side 2 m centered at $(3, 4, -5)$

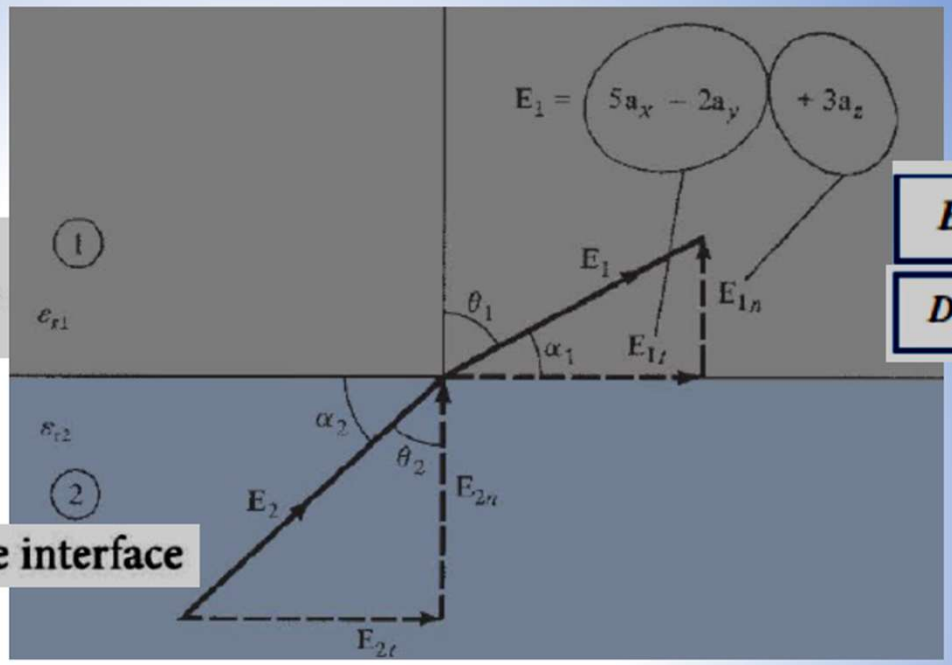


(a) $\mathbf{E}_2$ for $z \leq 0$

$$\mathbf{E}_{2t} = \mathbf{E}_{1t} = 5\mathbf{a}_x - 2\mathbf{a}_y$$

$$\mathbf{E}_{2n} = \frac{\varepsilon_{r1}}{\varepsilon_{r2}}\mathbf{E}_{1n} = \frac{4}{3}(3\mathbf{a}_z) = 4\mathbf{a}_z$$

$$\mathbf{E}_2 = \mathbf{E}_{2t} + \mathbf{E}_{2n}$$
$$= 5\mathbf{a}_x - 2\mathbf{a}_y + 4\mathbf{a}_z \text{ kV/m}$$



$$E_{\tan 1} = E_{\tan 2}$$

$$D_{N1} - D_{N2} = \rho_S$$

(b) The angles $\mathbf{E}_1$ and $\mathbf{E}_2$ make with the interface

$$\alpha_1 = 90 - \theta_1$$
$$\alpha_2 = 90 - \theta_2$$

$$\tan\theta_1 = \frac{E_{1t}}{E_{1n}} = \frac{\sqrt{29}}{3} = 1.795 \rightarrow \theta_1 = 60.9°$$

$$\alpha_1 = 29.1°$$

$$\tan\theta_2 = \frac{E_{2t}}{E_{2n}} = \frac{\sqrt{29}}{4} = 1.346 \rightarrow \theta_2 = 53.4°$$

$$\alpha_2 = 36.6°$$

Verify $\dfrac{\tan\theta_1}{\tan\theta_2} = \dfrac{\epsilon_1}{\epsilon_2}$

(c) The energy densities are given by $\mathbf{E}_1 = 5\mathbf{a}_x - 2\mathbf{a}_y + 3\mathbf{a}_z \text{ kV/m}$

$$w_{E1} = \frac{1}{2}\varepsilon_1|\mathbf{E}_1|^2 = \frac{1}{2} \times 4 \times \frac{10^{-9}}{36\pi} \times (25 + 4 + 9) \times 10^6$$
$$= 672\ \mu\text{J/m}^3$$

$$w_{E2} = \frac{1}{2}\varepsilon_2|\mathbf{E}_2|^2 = \frac{1}{2} \times 3 \times \frac{10^{-9}}{36\pi} (25 + 4 + 16) \times 10^6$$
$$= 597\ \mu\text{J/m}^3$$

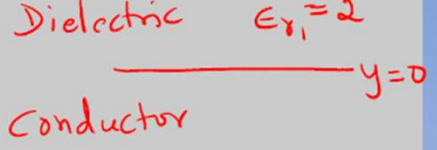
(d) The energy within a cube of side 2 m centered at $(3, 4, -5)$

(d) At the center $(3, 4, -5)$ of the cube of side 2 m, $z = -5 < 0$; that is, the cube is in region 2 with $2 \le x \le 4, 3 \le y \le 5, -6 \le z \le -4$. Hence

$$W_E = \int w_{E2}\, dv = \int_{x=2}^{4} \int_{y=3}^{5} \int_{z=-6}^{-4} w_{E2}\, dz\, dy\, dz$$
$$= w_{E2}(2)(2)(2)$$
$$= 597 \times 8\ \mu\text{J} = 4.776\ \text{mJ}$$

Region $y < 0$ consists of a perfect conductor while region $y > 0$ is a dielectric medium ($\varepsilon_{1r} = 2$) as in ~~Figure 5.16~~. If there is a surface charge of 2 nC/m^2 on the conductor, determine **E** and **D** at

(a) $A(3, -2, 2)$
(b) $B(-4, 1, 5)$



(a) Point $A(3, -2, 2)$ is in the conductor since $y = -2 < 0$ at A. Hence,

$$\mathbf{E} = \mathbf{0} = \mathbf{D}$$

(b) Point $B(-4, 1, 5)$ is in the dielectric medium since $y = 1 > 0$ at B.

$$D_n = \rho_S = 2\,\text{nC/m}^2$$

$$\mathbf{D} = 2\mathbf{a}_y\,\text{nC/m}^2$$

$$\mathbf{E} = \frac{\mathbf{D}}{\varepsilon_o \varepsilon_r} = 2 \times 10^{-9} \times \frac{36\pi}{2} \times 10^9 \mathbf{a}_y = 36\pi \mathbf{a}_y$$

$$= 113.1\,\mathbf{a}_y\,\text{V/m}$$

Dr. Sampath

D5.9. Let Region 1 ($z < 0$) be composed of a uniform dielectric material for which $\epsilon_r = 3.2$, while Region 2 ($z > 0$) is characterized by $\epsilon_r = 2$. Let $\mathbf{D}_1 = -30\mathbf{a}_x + 50\mathbf{a}_y + 70\mathbf{a}_z$ nC/m^2 and find: (*a*) D_{N1}; (*b*) $\mathbf{D}_{t1}$; (*c*) D_{t1}; (*d*) D_1; (*e*) θ_1; (*f*) $\mathbf{P}_1$.

Ans. (*a*) 70 nC/m^2; (*b*) $-30\mathbf{a}_x + 50\mathbf{a}_y$ nC/m^2; (*c*) 58.3 nC/m^2; (*d*) 91.1 nC/m^2; (*e*) 39.8°; (*f*) $-20.6\mathbf{a}_x + 34.4\mathbf{a}_y + 48.1\mathbf{a}_z$ nC/m^2

D5.10. Continue Problem D5.9 by finding: (*a*) $\mathbf{D}_{N2}$; (*b*) $\mathbf{D}_{t2}$; (*c*) $\mathbf{D}_2$; (*d*) $\mathbf{P}_2$; (*e*) θ_2.

Ans. (*a*) $70\mathbf{a}_z$ nC/m^2; (*b*) $-18.75\mathbf{a}_x + 31.25\mathbf{a}_y$ nC/m^2; (*c*) $-18.75\mathbf{a}_x + 31.25\mathbf{a}_y + 70\mathbf{a}_z$ nC/m^2; (*d*) $-9.38\mathbf{a}_x + 15.63\mathbf{a}_y + 35\mathbf{a}_z$ nC/m^2; (*e*) 27.5°

P = polarization

We locate a slab of Teflon in the region $0 \leq x \leq a$, and assume free space where $x < 0$ and $x > a$. Outside the Teflon there is a uniform field $\mathbf{E}_{out} = E_0\mathbf{a}_x$ V/m. We seek values for **D**, **E**, and **P** everywhere.

Solution. The dielectric constant of the Teflon is 2.1, and thus the electric susceptibility is 1.1.

Outside the slab, we have immediately $\mathbf{D}_{out} = \epsilon_0 E_0 \mathbf{a}_x$. Also, as there is no dielectric material there, $\mathbf{P}_{out} = 0$. Now, any of the last four or five equations will enable us to relate the several fields inside the material to each other. Thus

$$\mathbf{D}_{in} = 2.1\epsilon_0 \mathbf{E}_{in} \qquad (0 \leq x \leq a)$$

$$\mathbf{P}_{in} = 1.1\epsilon_0 \mathbf{E}_{in} \qquad (0 \leq x \leq a)$$

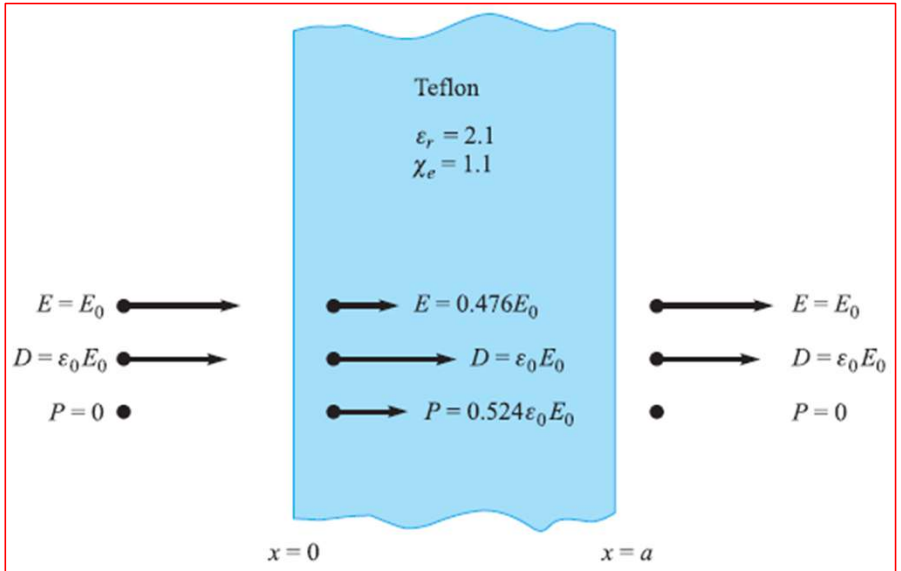


Figure 5.12 A knowledge of the electric field external to the dielectric enables us to find the remaining external fields first and then to use the continuity of normal **D** to begin finding the internal fields.

Complete Example 5.4 by finding the fields within the Teflon ($\epsilon_r = 2.1$), given the uniform external field $\mathbf{E}_{out} = E_0\mathbf{a}_x$ in free space.

Solution. We recall that we had a slab of Teflon extending from $x = 0$ to $x = a$, as shown in Figure 5.12, with free space on both sides of it and an external field $\mathbf{E}_{out} = E_0\mathbf{a}_x$. We also have $\mathbf{D}_{out} = \epsilon_0 E_0\mathbf{a}_x$ and $\mathbf{P}_{out} = 0$.

Inside, the continuity of D_N at the boundary allows us to find that $\mathbf{D}_{in} = \mathbf{D}_{out} = \epsilon_0 E_0\mathbf{a}_x$. This gives us $\mathbf{E}_{in} = \mathbf{D}_{in}/\epsilon = \epsilon_0 E_0\mathbf{a}_x/(\epsilon_r\epsilon_0) = 0.476E_0\mathbf{a}_x$. To get the polarization field in the dielectric, we use $\mathbf{D} = \epsilon_0\mathbf{E} + \mathbf{P}$ and obtain

$$\mathbf{P}_{in} = \mathbf{D}_{in} - \epsilon_0\mathbf{E}_{in} = \epsilon_0 E_0\mathbf{a}_x - 0.476\epsilon_0 E_0\mathbf{a}_x = 0.524\epsilon_0 E_0\mathbf{a}_x$$

Summarizing then gives

$$\mathbf{D}_{in} = \epsilon_0 E_0\mathbf{a}_x \qquad (0 \le x \le a)$$
$$\mathbf{E}_{in} = 0.476E_0\mathbf{a}_x \qquad (0 \le x \le a)$$
$$\mathbf{P}_{in} = 0.524\epsilon_0 E_0\mathbf{a}_x \qquad (0 \le x \le a)$$

A practical problem most often does not provide us with a direct knowledge of the field on either side of the boundary. The boundary conditions must be used to help us determine the fields on both sides of the boundary from the other information that is given.

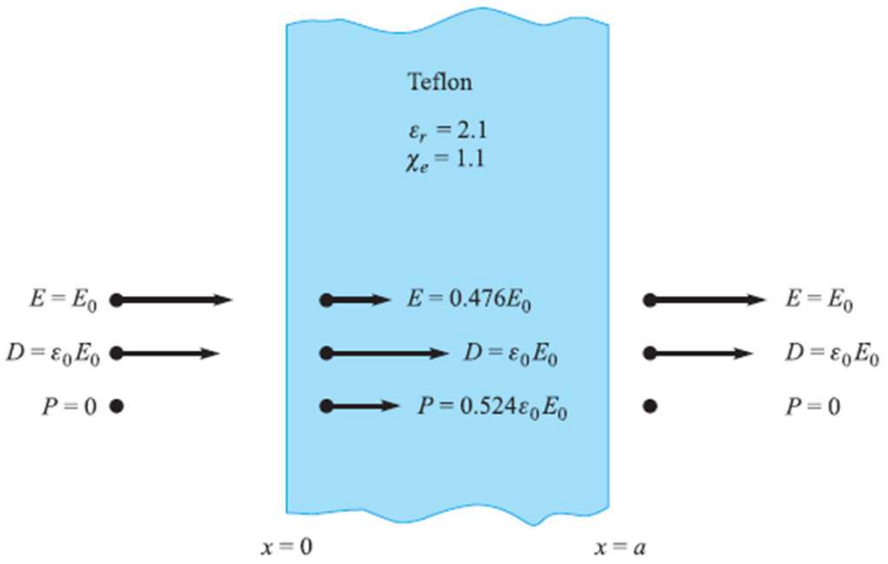


Figure 5.12 A knowledge of the electric field external to the dielectric enables us to find the remaining external fields first and then to use the continuity of normal D to begin finding the internal fields.

Current and Current Density

- Electric charges in motion constitute a *current*.
- The unit of current is the ampere (A), defined as a rate of movement of charge passing a given reference point (or crossing a given reference plane).

$$I = \frac{dQ}{dt}$$

- Current is *defined* as the motion of positive charges, although conduction in metals takes place through the motion of electrons.
- A new quantity, current density J is defined. It is measured in amperes per square meter (A/m^2).
- Through integration, the total current is obtained:

$$I = \int_S \mathbf{J} \cdot d\mathbf{S}$$

D5.1. Given the vector current density $\mathbf{J} = 10\rho^2 z\mathbf{a}_\rho - 4\rho\cos^2\phi\,\mathbf{a}_\phi$ mA/m^2: (*a*) find the current density at $P(\rho = 3, \phi = 30°, z = 2)$; (*b*) determine the total current flowing outward through the circular band $\rho = 3, 0 < \phi < 2\pi, 2 < z < 2.8$.

Ans. (*a*) $180\mathbf{a}_\rho - 9\mathbf{a}_\phi$ mA/m^2; (*b*) 3.26 A

Hint: pdϕdz **aρ**

Current and Current Density

$$\mathbf{J} = \rho_v \mathbf{v}$$

■ J = ρ_vv is then called *convection current density*.

V = velocity of charge density

Also, $\mathbf{J} = \sigma \mathbf{E}$ σ = conductivity

$$I = \oint_S J \cdot \mathrm{d}\boldsymbol{S} = -\frac{dQ_i}{dt}$$

- **The Integral Form of the Continuity Equation**

$$\int_{\text{vol}} (\nabla \cdot \mathbf{J}) dv = -\frac{d}{dt} \int_{\text{vol}} \rho_v dv$$

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}$$

- **The Differential Form (Point Form) of the Continuity Equation**